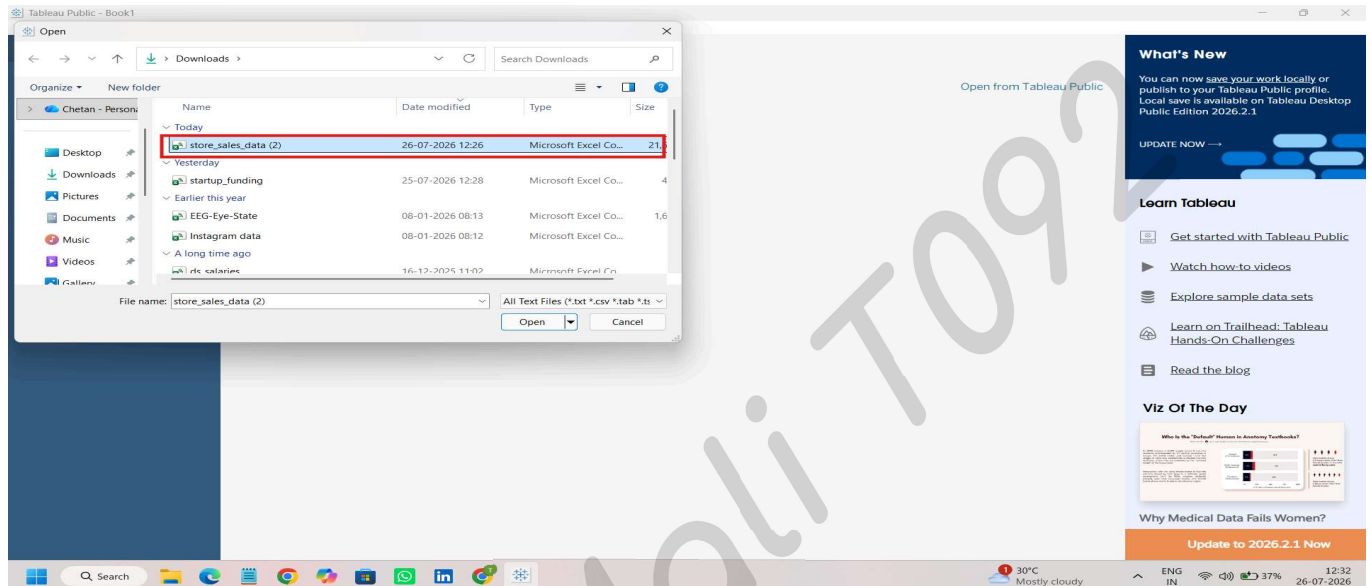


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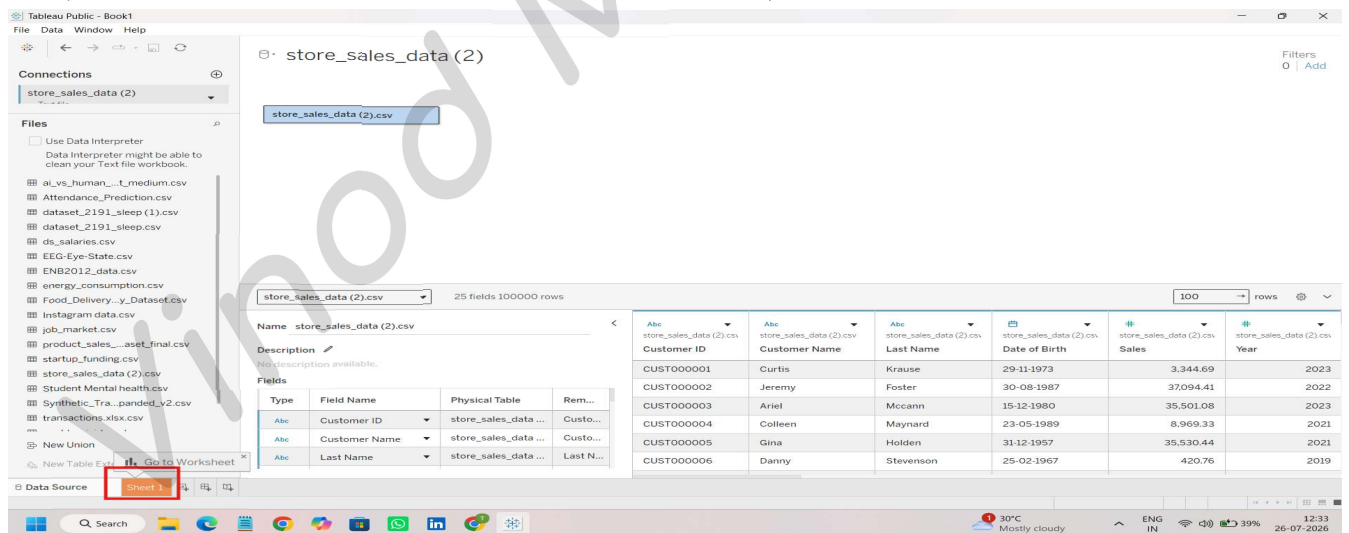
Aim:- Performing Data Analysis using Calculations in Tableau

A) Create basic calculated fields using formulas

Step 1:- Open Tableau And Connect The Dataset



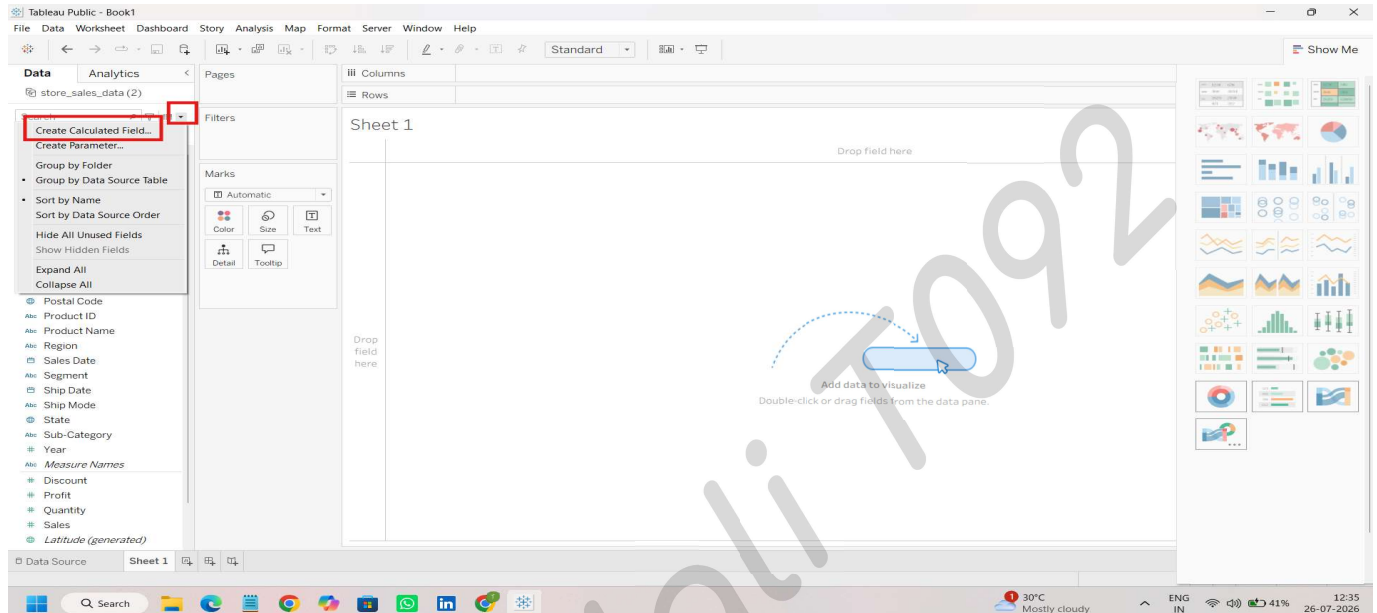
Step 2:- Click On Sheet 1 To Enter In Workspace



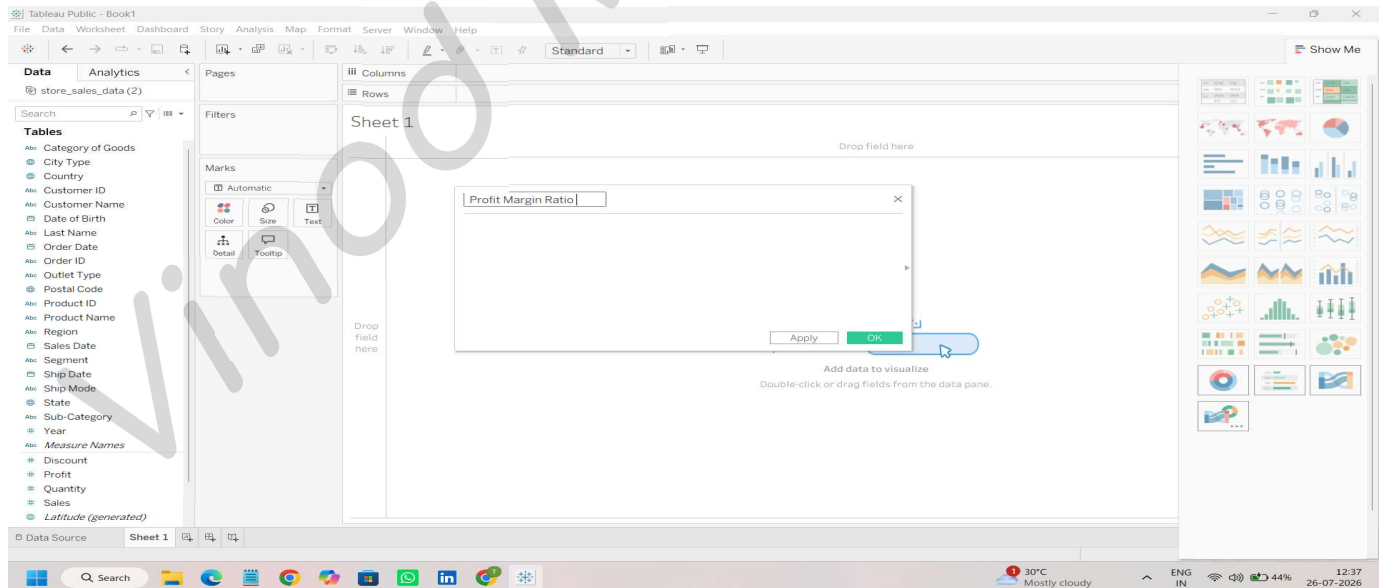
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Roll No :- T092

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Step 3:- In Data Pane Which is Located In Left Side , Pane. Right-click anywhere in the blank space of the pane and select Create Calculated Field



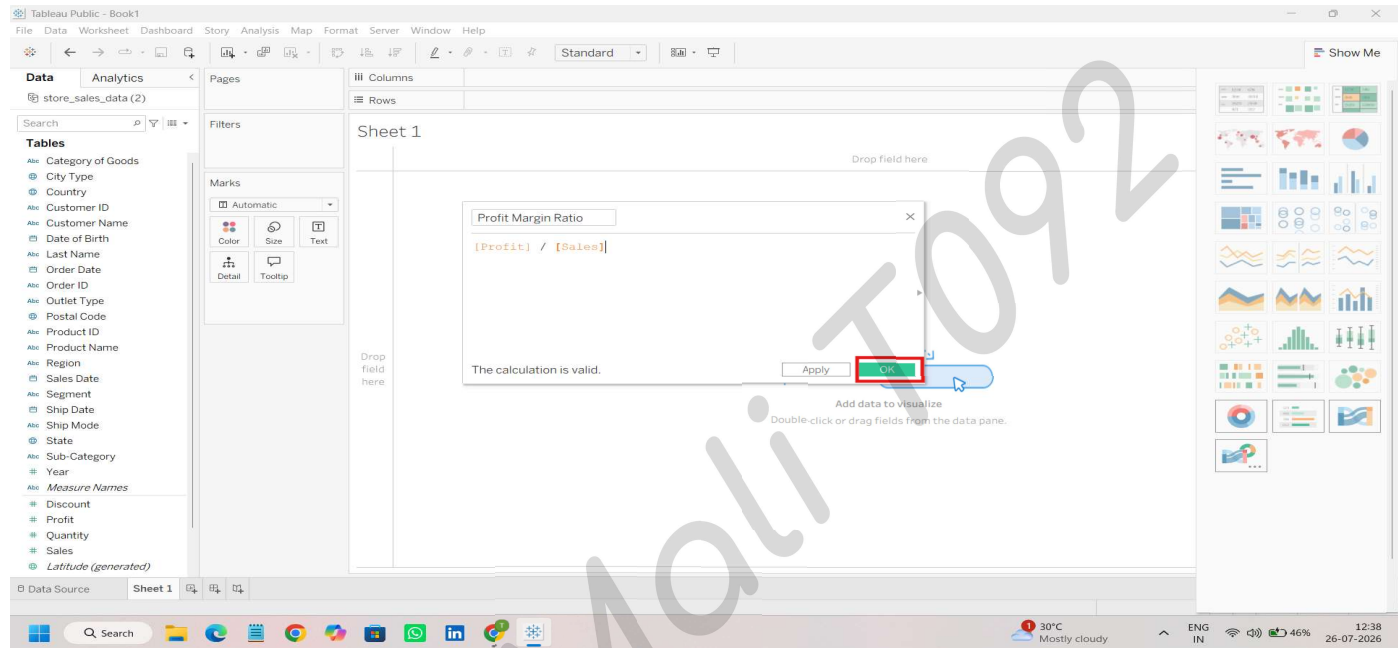
Step 4:- Calculation Windows Will pop Up, Clear The Default Name and Name The Field Profit Margin Ratio



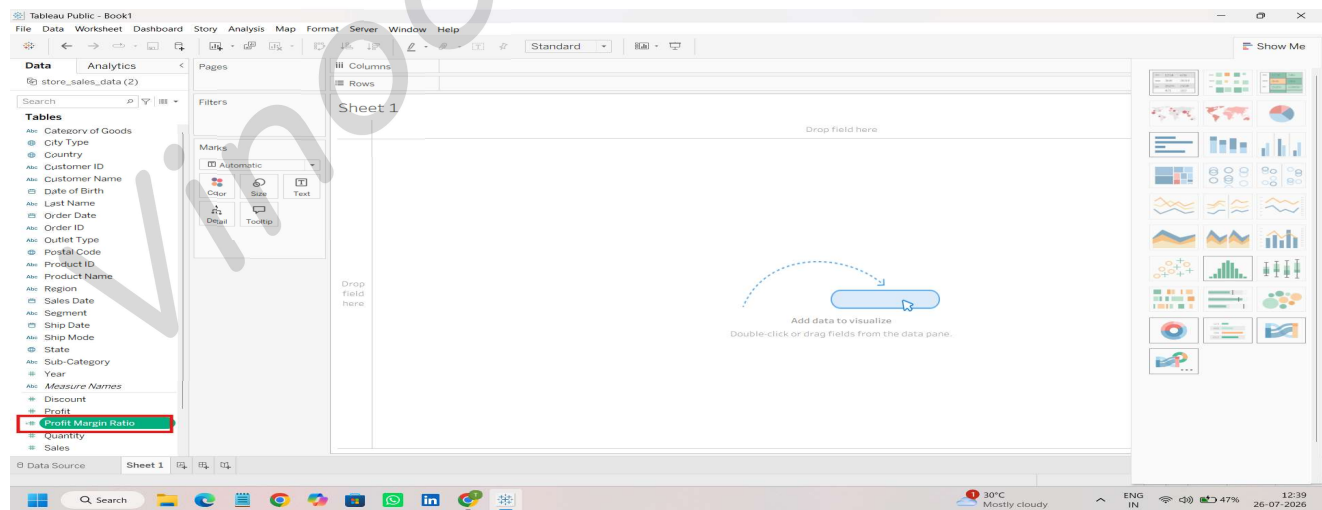
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Step 5:- In the main workspace box, enter the following basic mathematical formula: $[\text{Profit}] / [\text{Sales}]$. Look at the bottom-left corner of the editor box. It should say "The calculation is valid." Click OK.



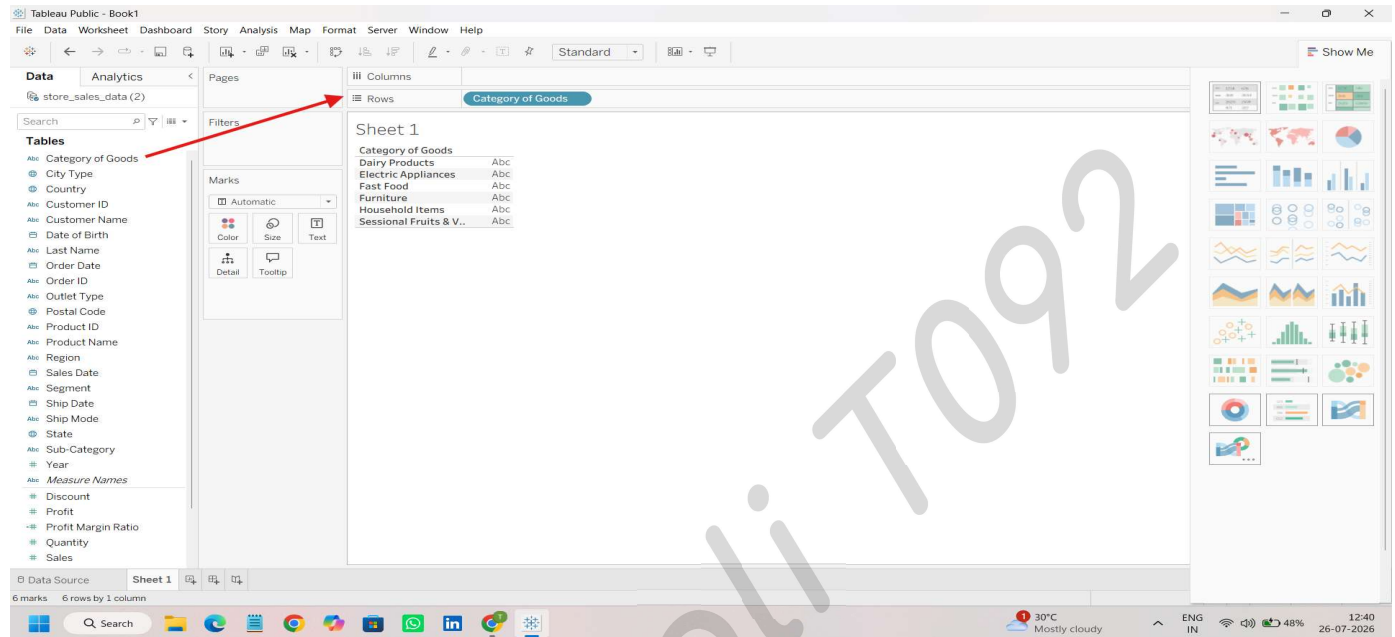
Step 6:- Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.



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Step 7:- Drag the Category of Goods dimension from your Data Pane and drop it onto the Rows shelf.



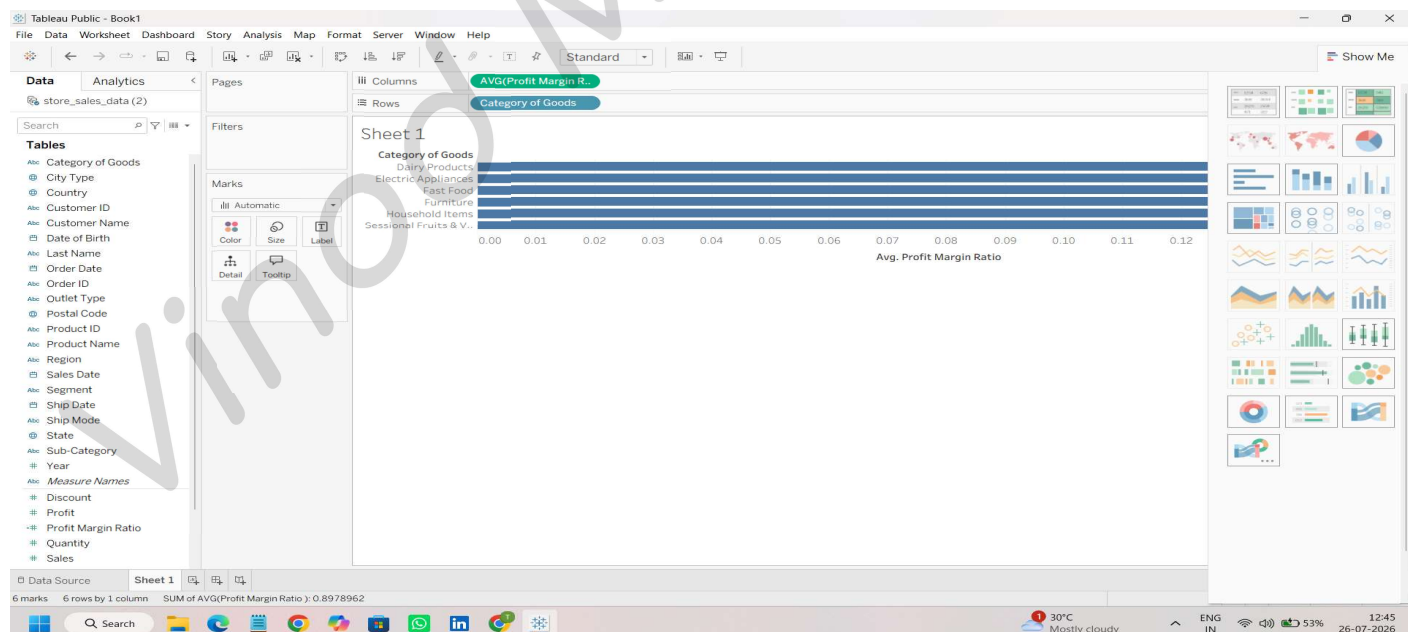
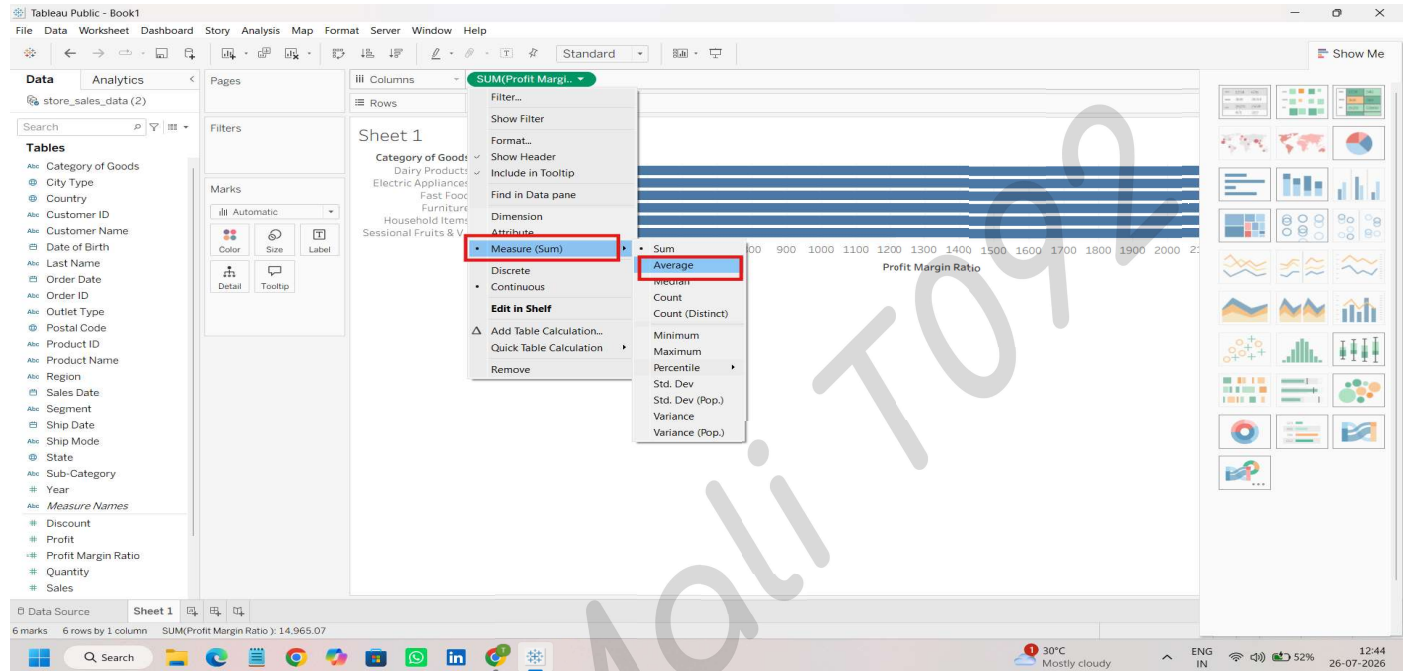
Step 8:- Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



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Step 9:- Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.

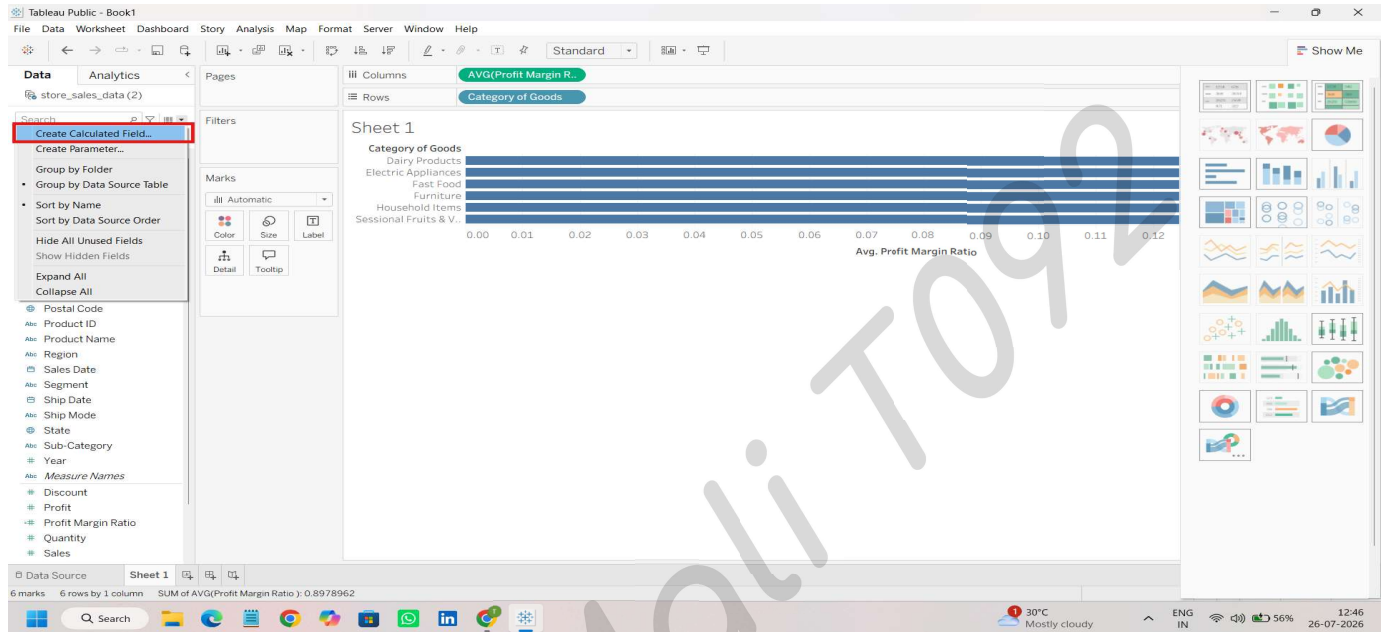


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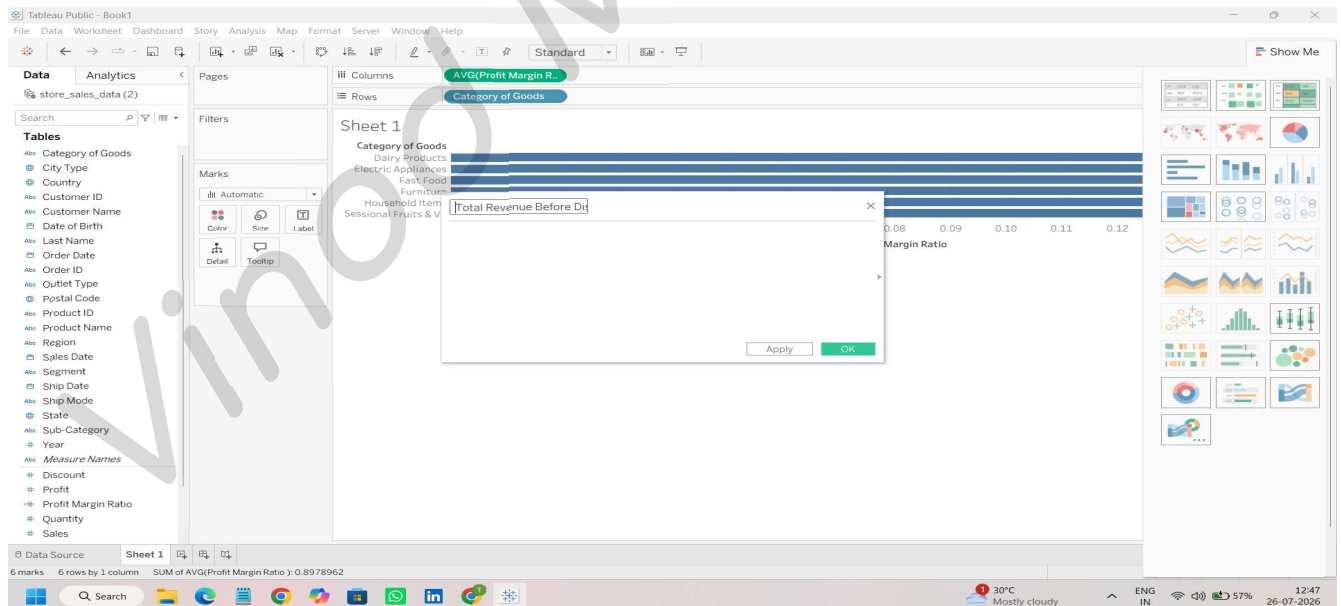
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B) Perform row-level calculations

Step 1:- Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...



Step 2:- Name the field: Total Revenue Before Discount.

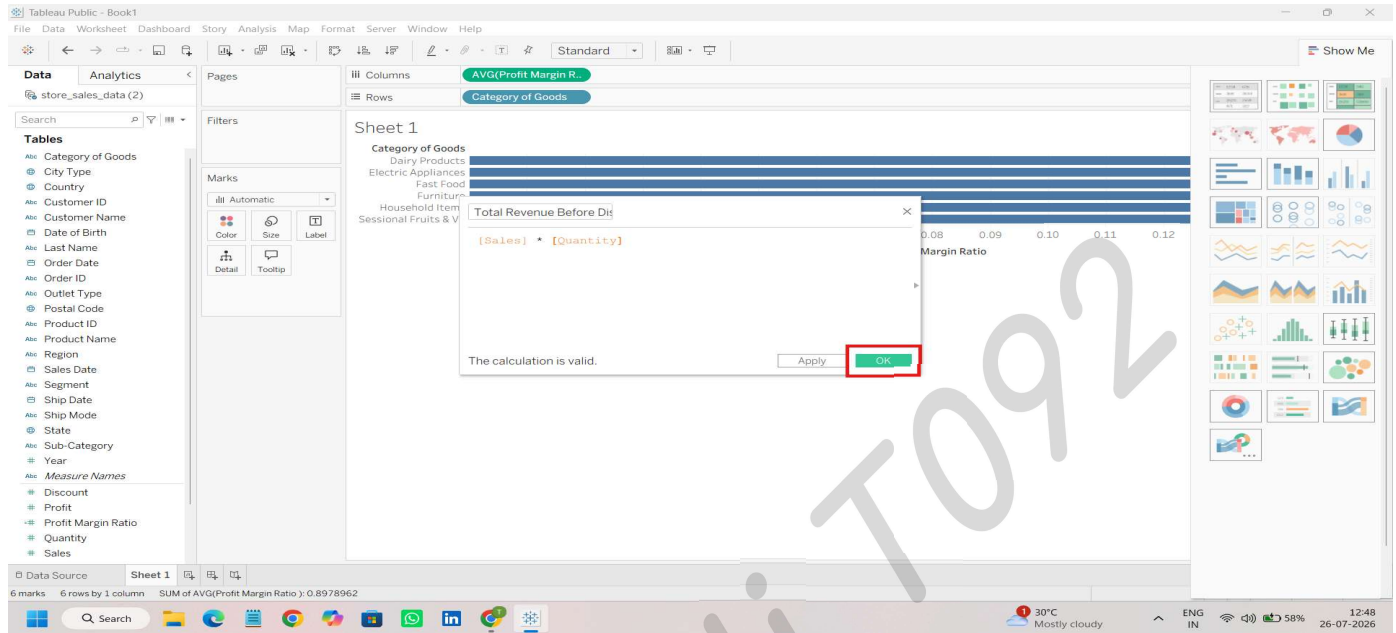


Step 3:- In the calculation formula canvas area, input the following expression: $[Sales] * [Quantity]$ Verify the message at the bottom reads "The calculation is valid" and click OK.

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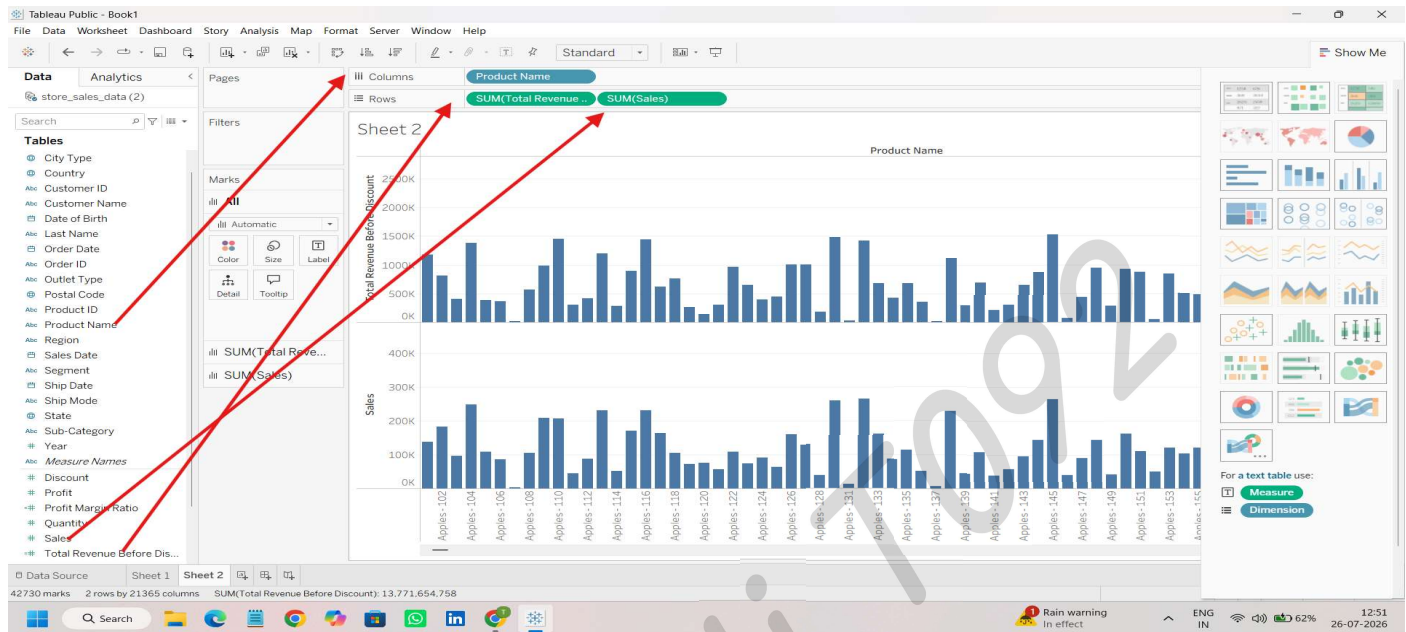
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Step 4:-Click On Sheet 2 And Drag the Product Name dimension from the Data Pane and drop it into the Rows shelf. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

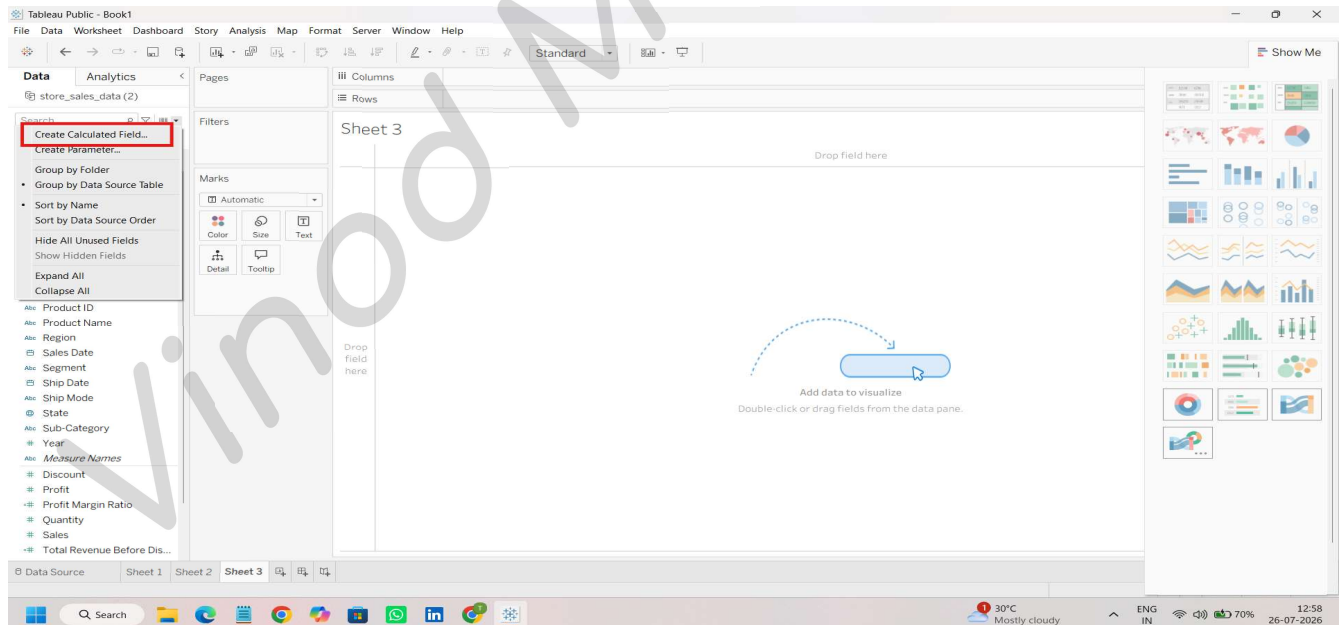
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C) Perform aggregate-level calculations

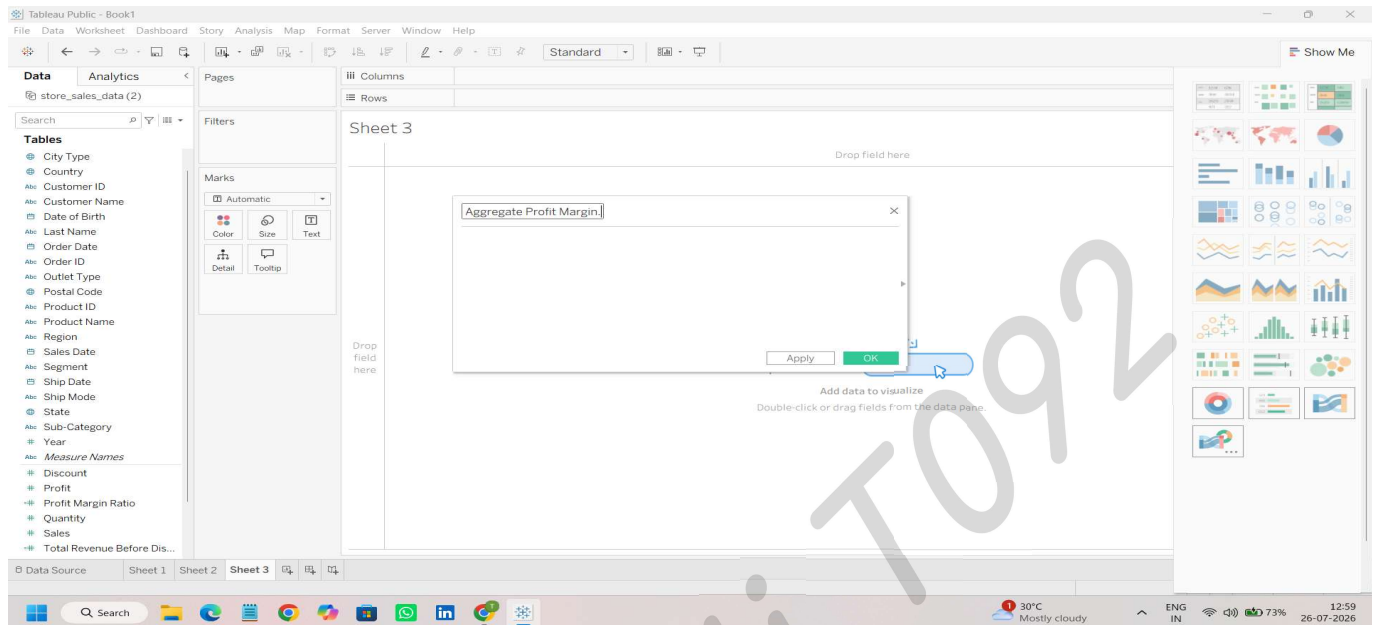
Step 1:- Click On New Worksheet And Right-click in the blank white space of your Data Pane and select Create Calculated Field...



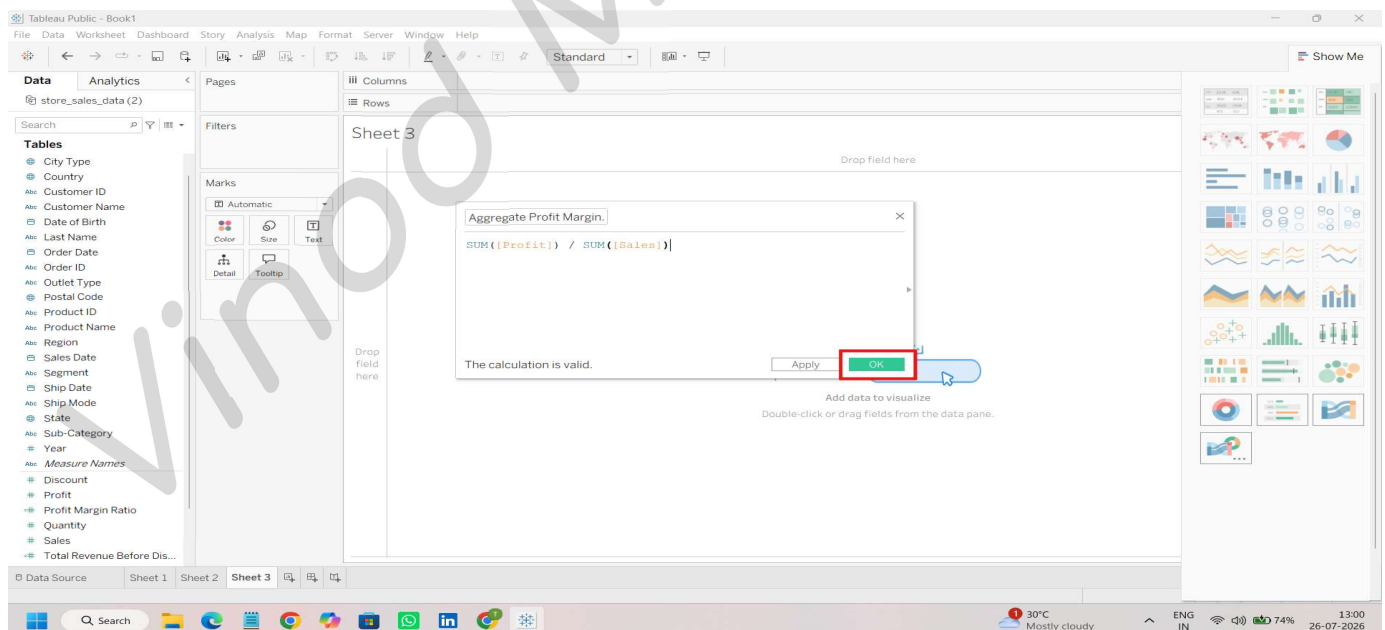
Step 2:- Name the field: Aggregate Profit Margin.

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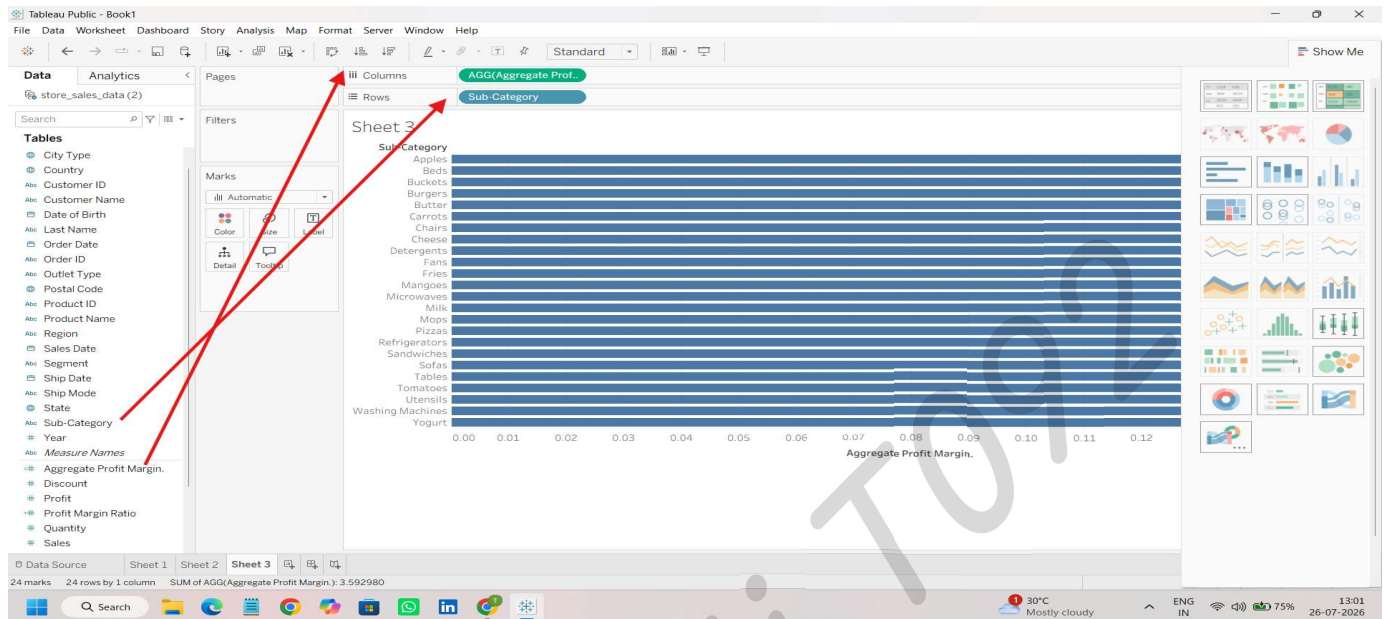
Step 3:- In the calculation formula window, type the following expression:
 $\text{SUM}([\text{Profit}]) / \text{SUM}([\text{Sales}])$ and Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



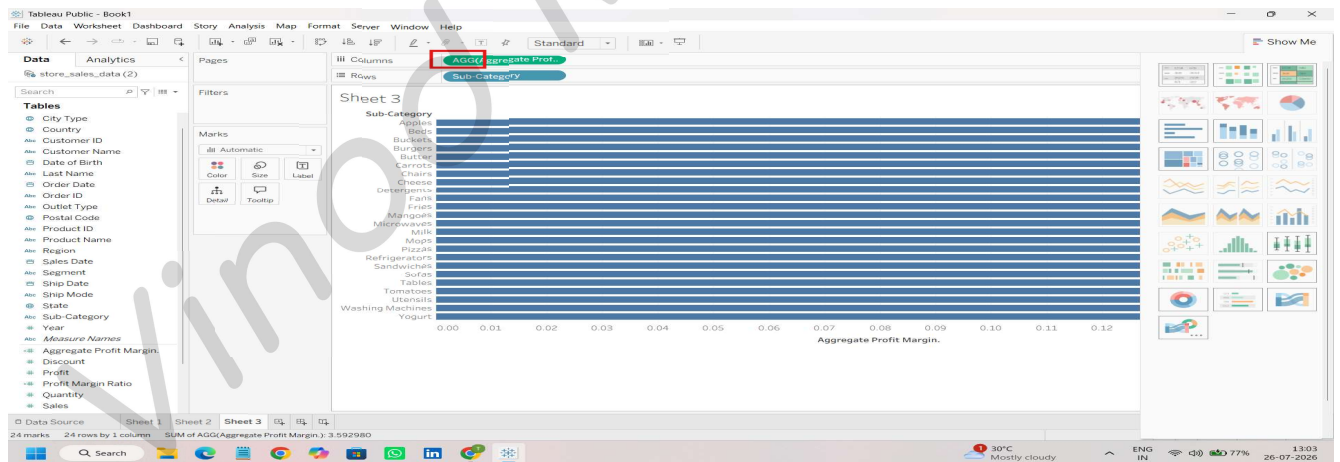
Step 4:- Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.

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Step 5:- Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.

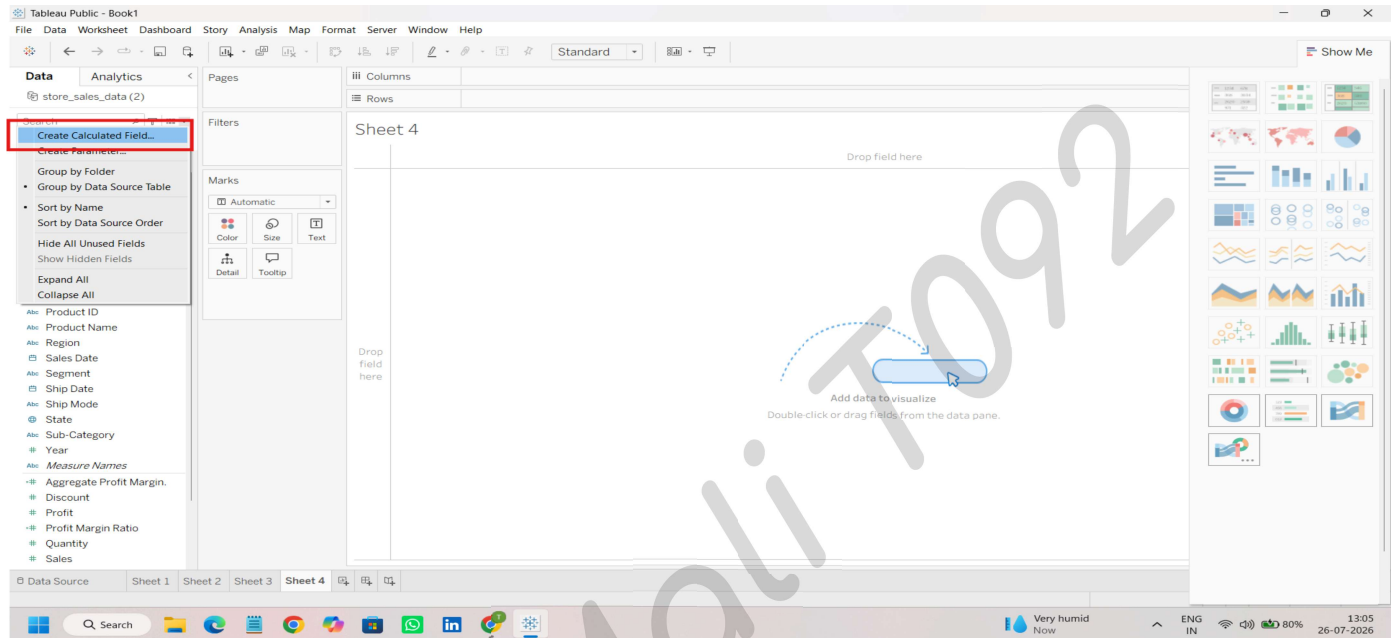


D) Implement FIXED Level of Detail (LOD) calculations

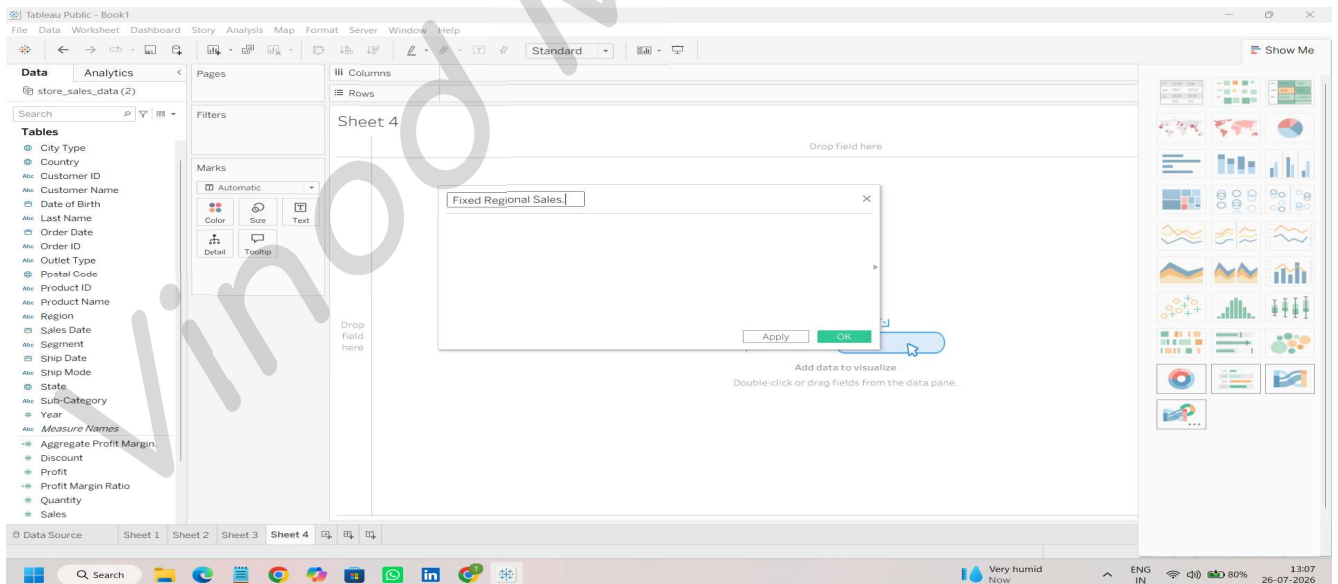
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Step 1:- Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



Step 2:- Name the field: Fixed Regional Sales.

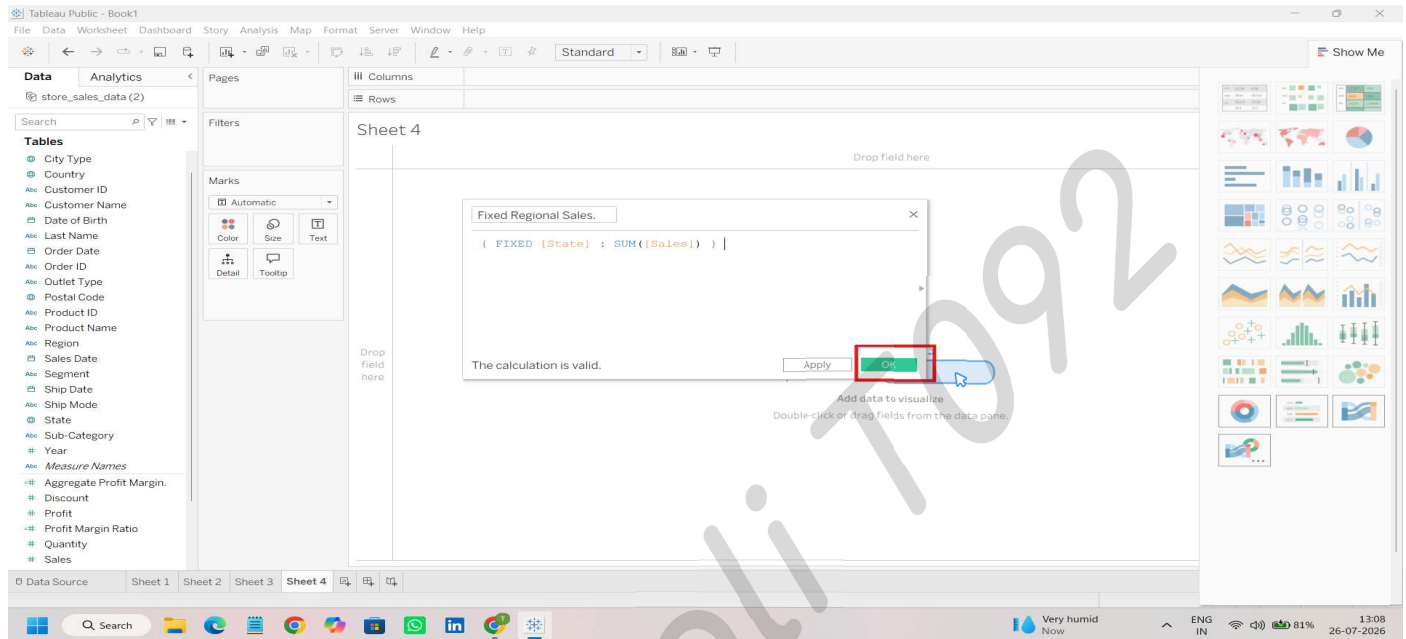


Step 3:- In the calculation formula window, type the following expression using curly braces {}: { FIXED [State] : SUM([Sales]) } Look

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Roll No :- T092

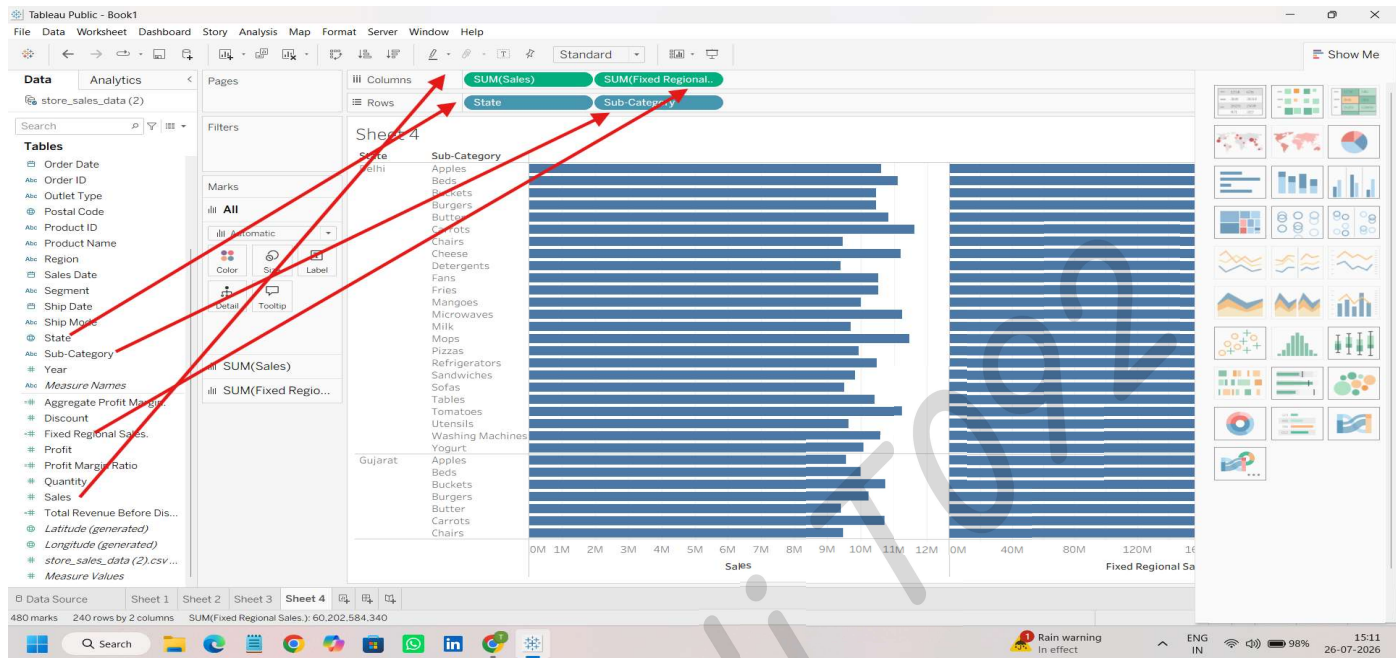
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at the bottom left to ensure it reads "The calculation is valid" and click OK.

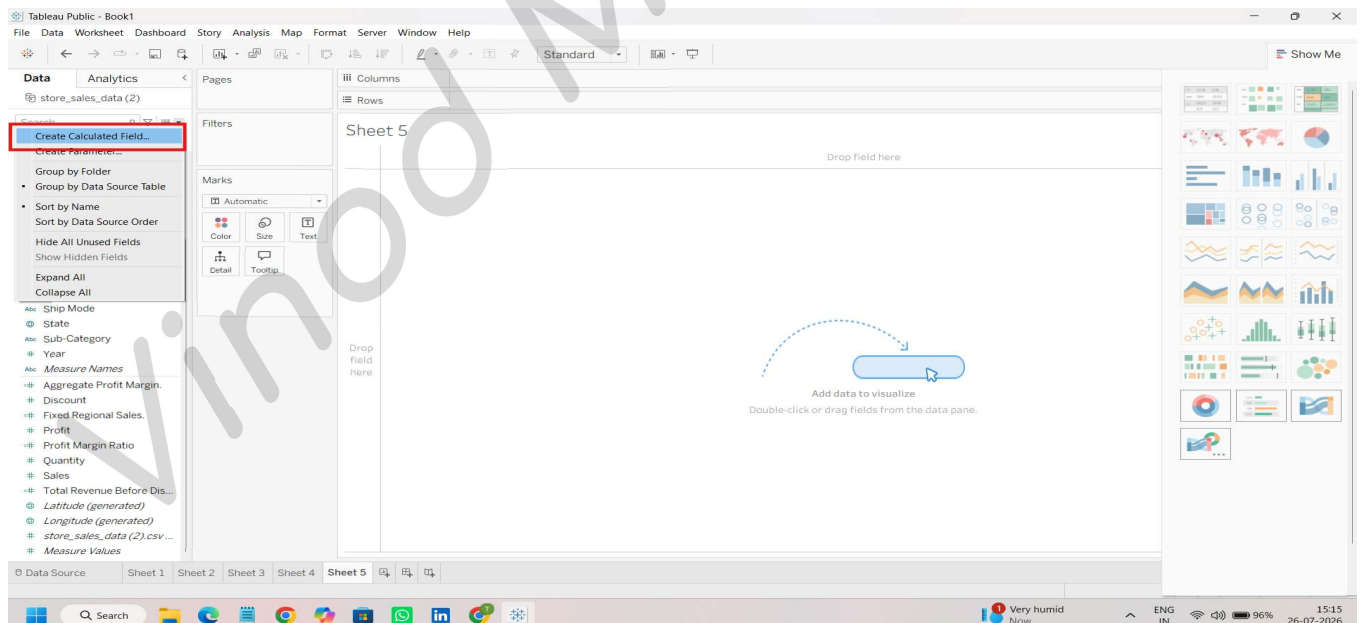


Step 4:- Drag the State dimension from the Data Pane and drop it onto the Rows shelf. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State. Drag the row Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state). Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the row Sales pill.

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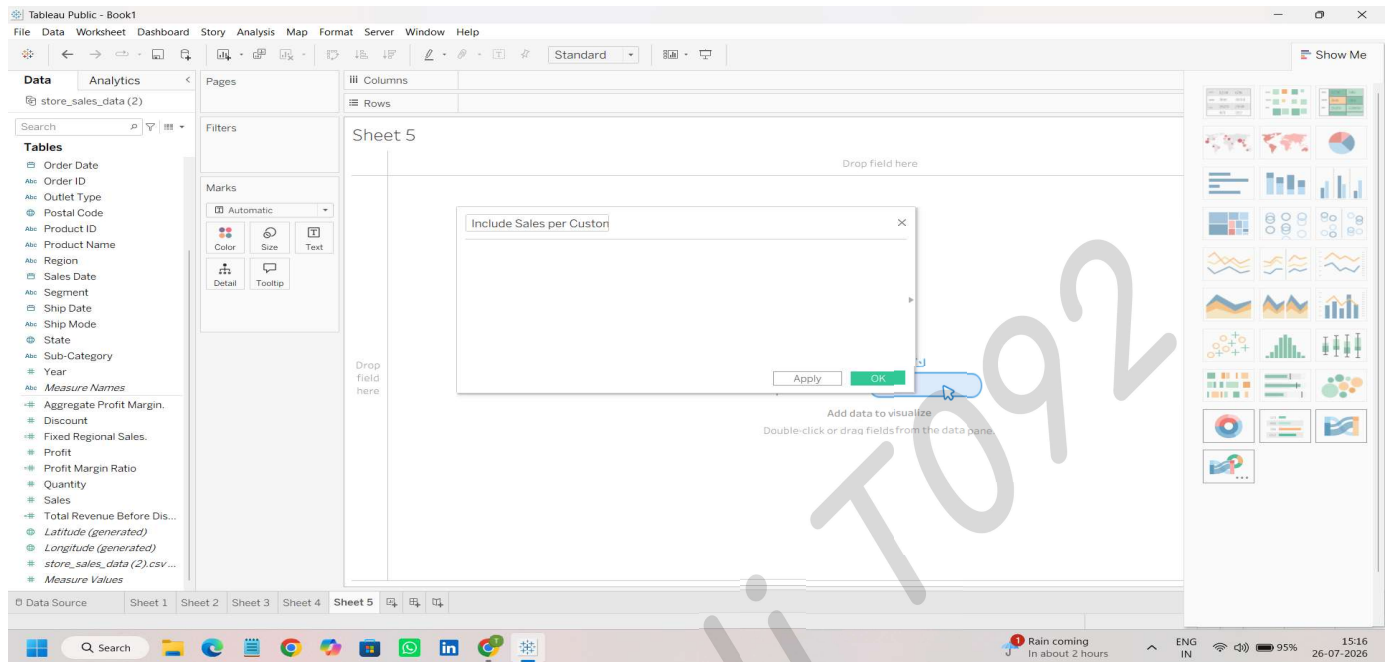


E) Use INCLUDE and EXCLUDE LOD expressions
Step 1:- Right-click in the blank white space of your Data Pane and select Create Calculated Field...

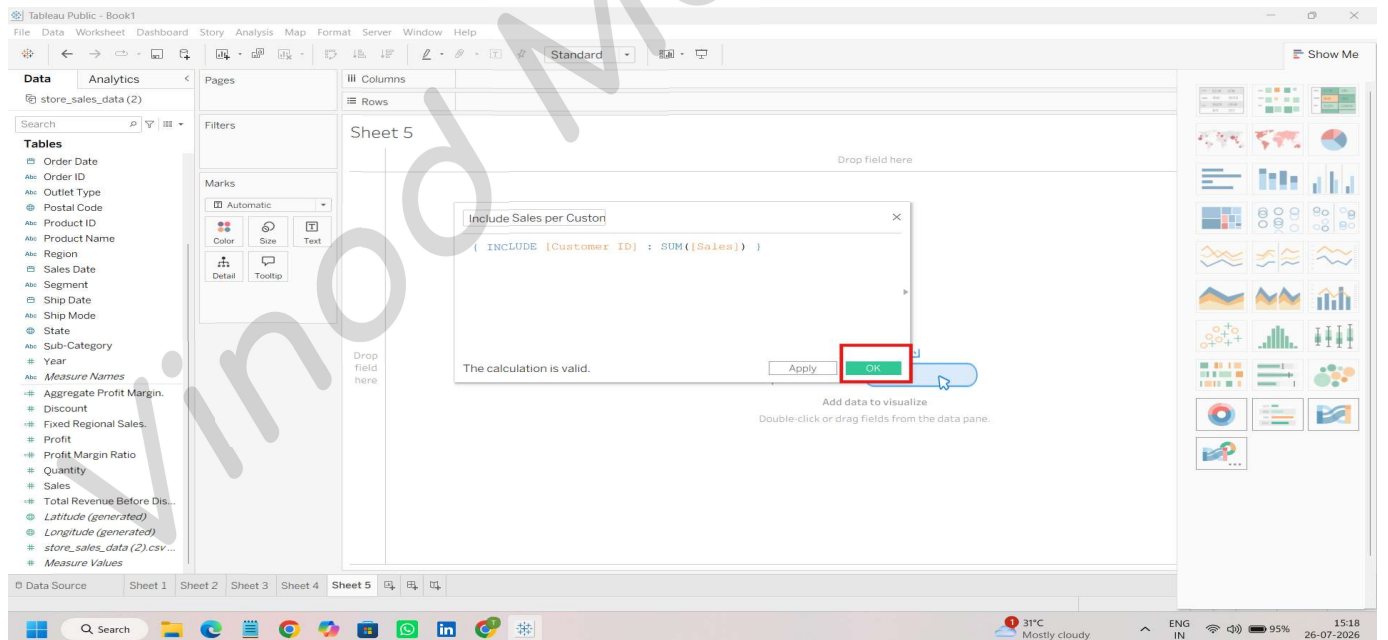


Step 2:- Name the field: Include Sales per Customer.
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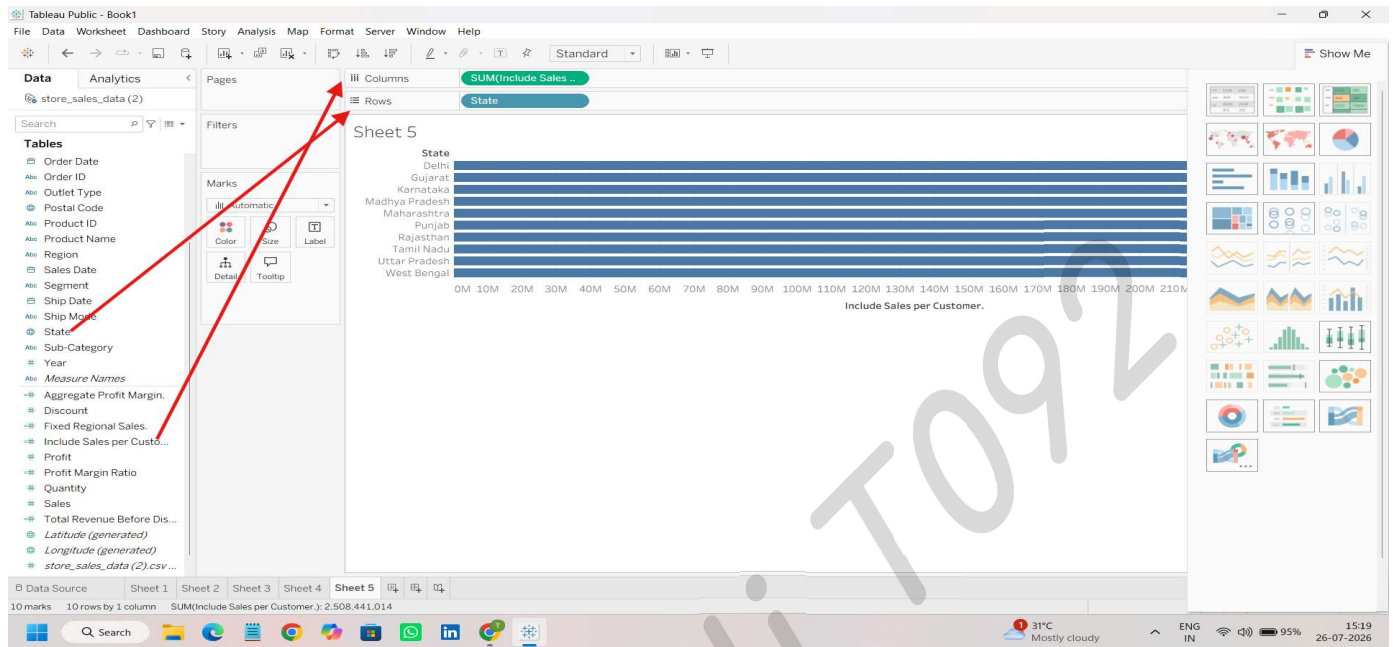
Step 3:- In the calculation formula canvas area, input the following expression: { INCLUDE [Customer ID] : SUM([Sales]) } and Click OK



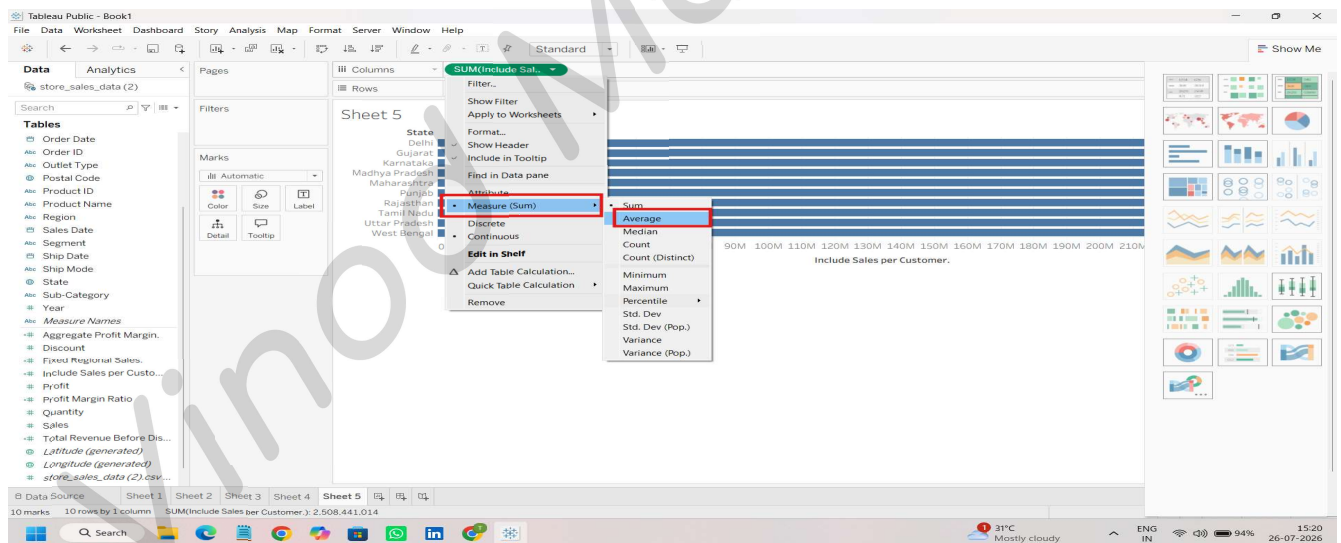
Step 4:- Drag the State dimension from the Data Pane and drop it onto the Rows shelf. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.

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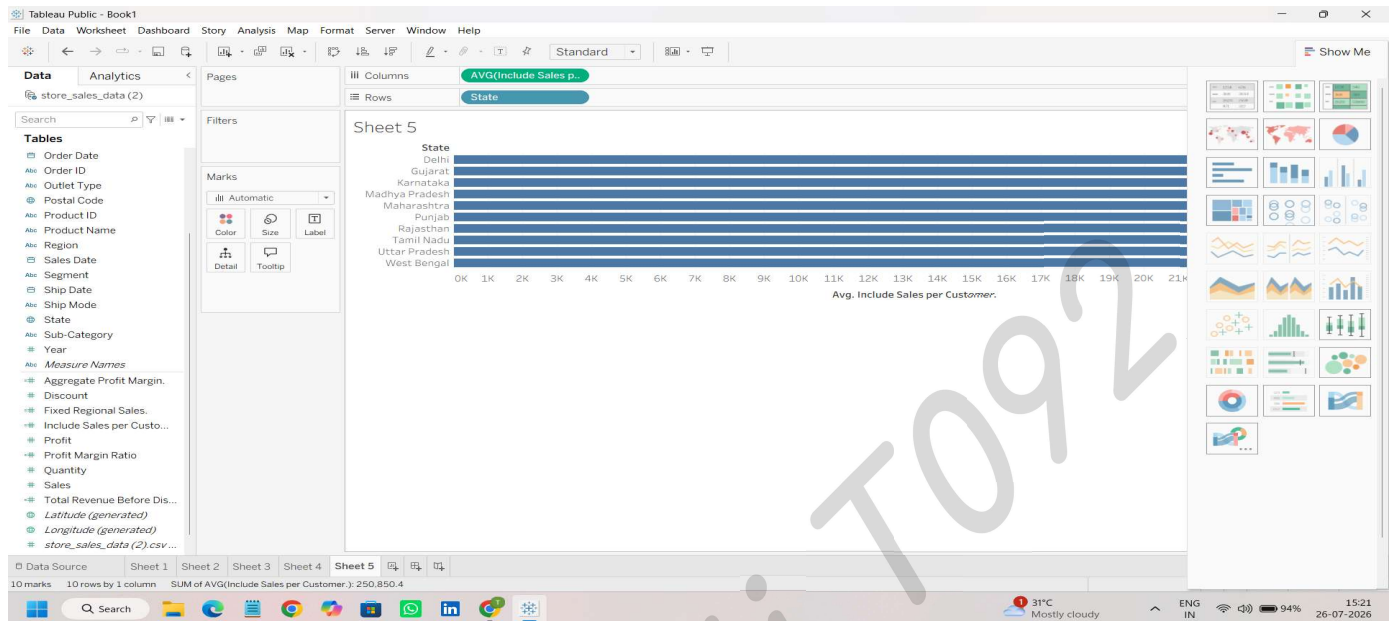


Step 5:- Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.

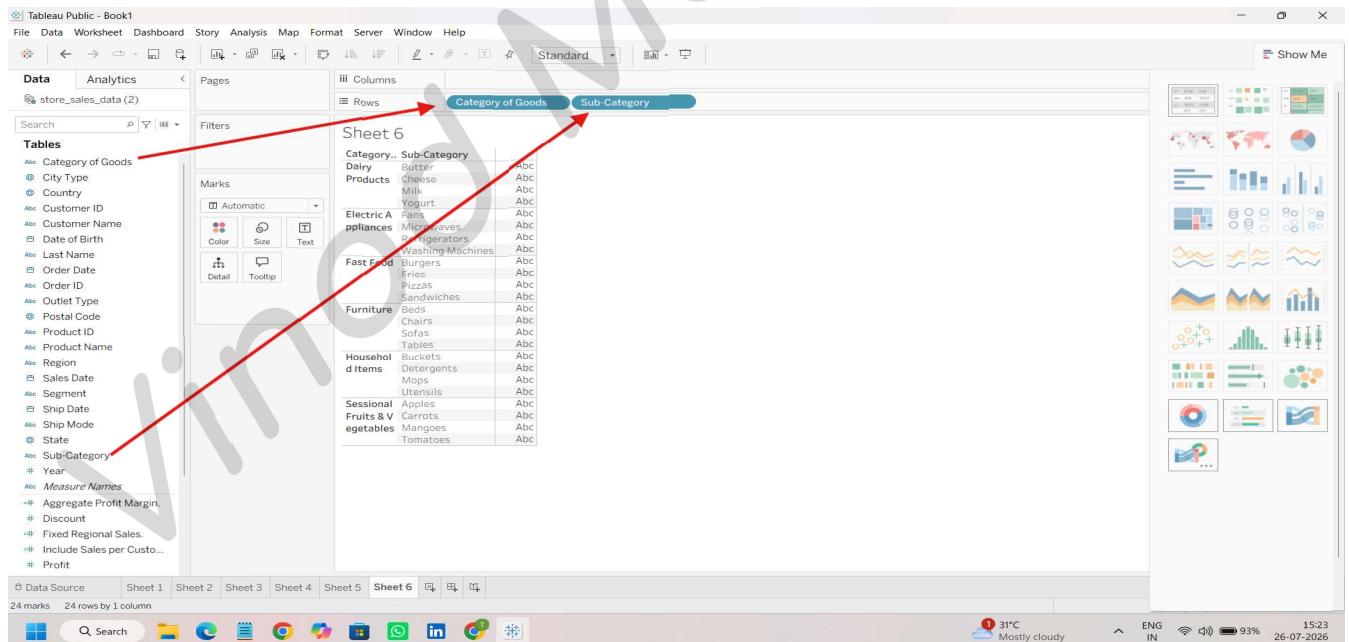


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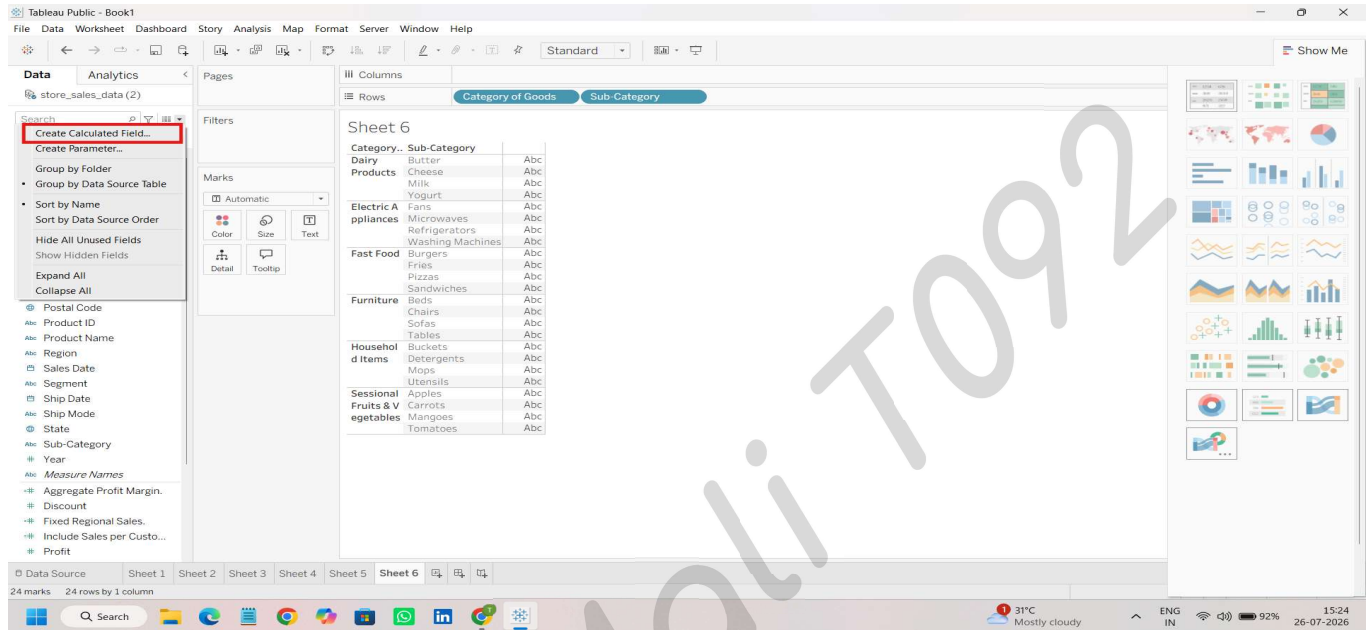
Step 6:- Open a new blank worksheet. Drag Category of Goods onto the Rows shelf. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).



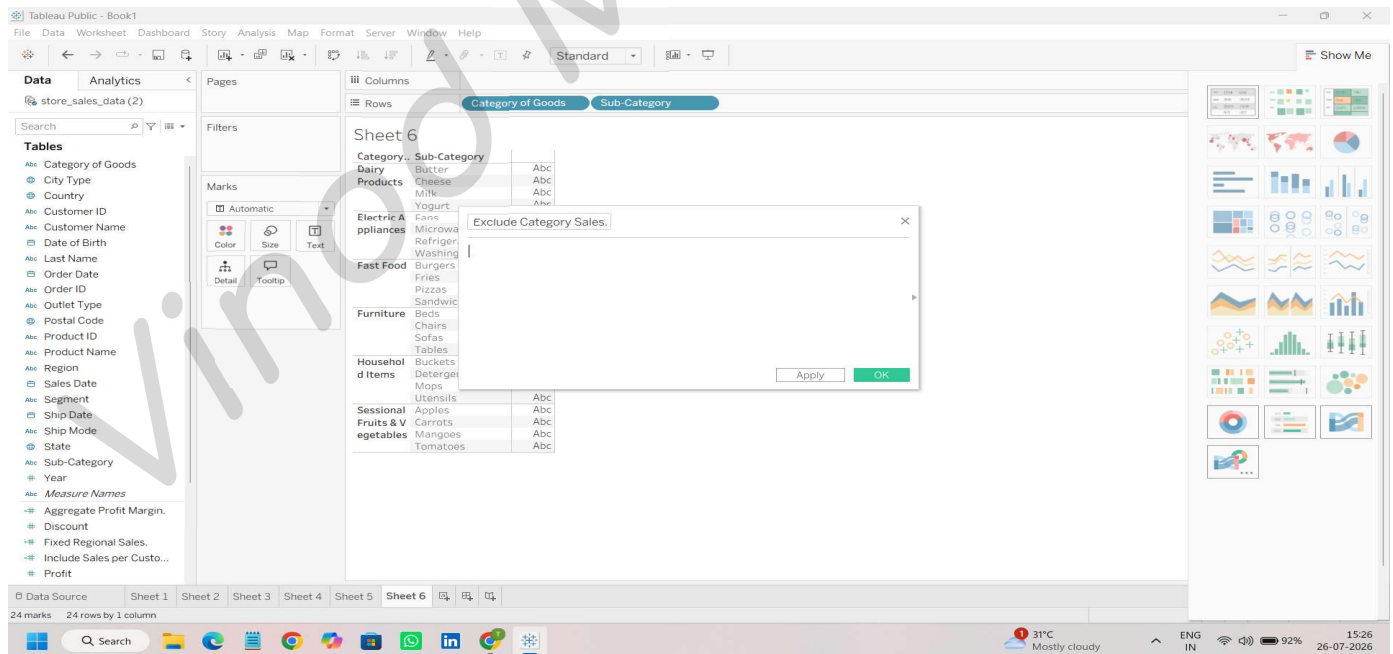
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Step 7:- Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



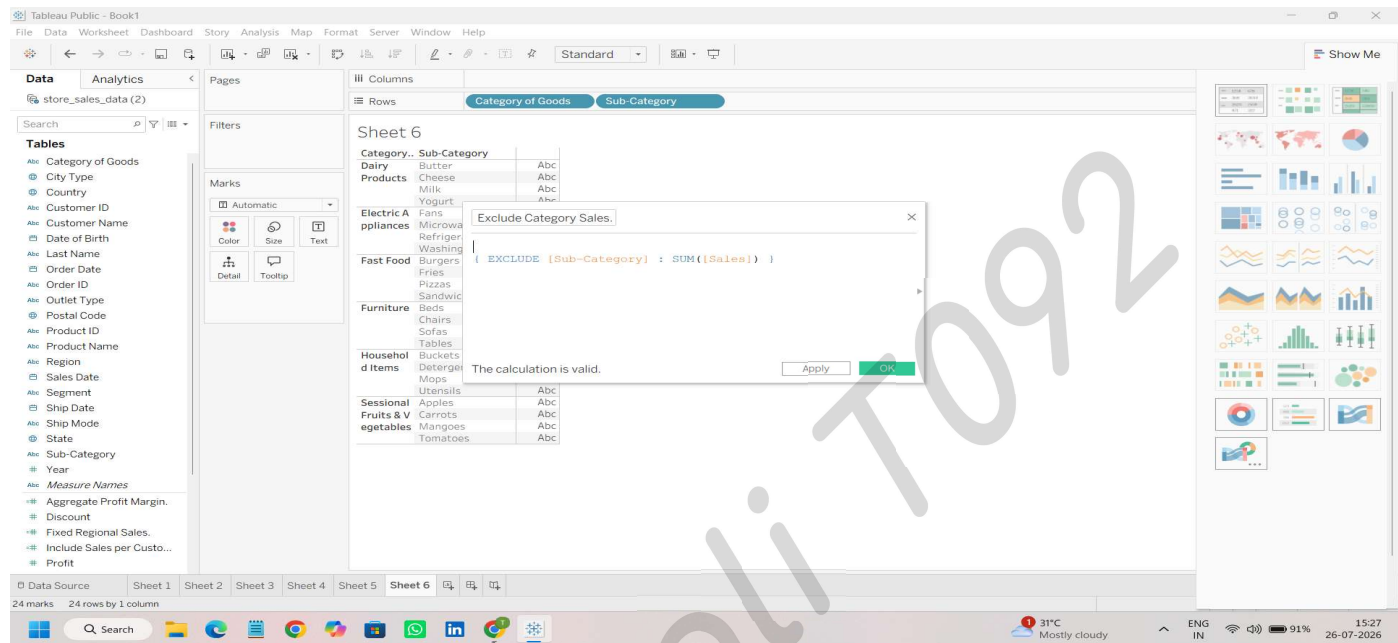
Step 8:- Name the field: Exclude Category Sales.



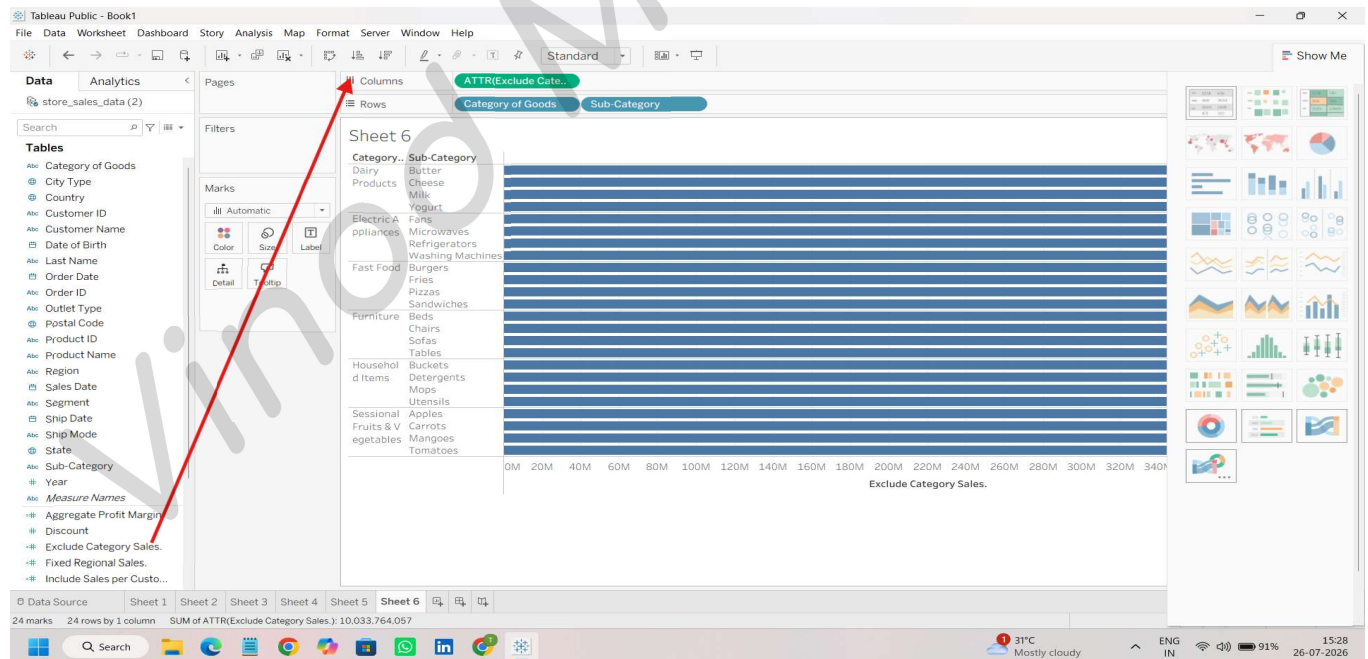
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Step 9:- Input the following expression: { EXCLUDE [Sub-Category] :
SUM([Sales]) } And Click OK



Step 10:- Drag This New Field Into Columns Itself

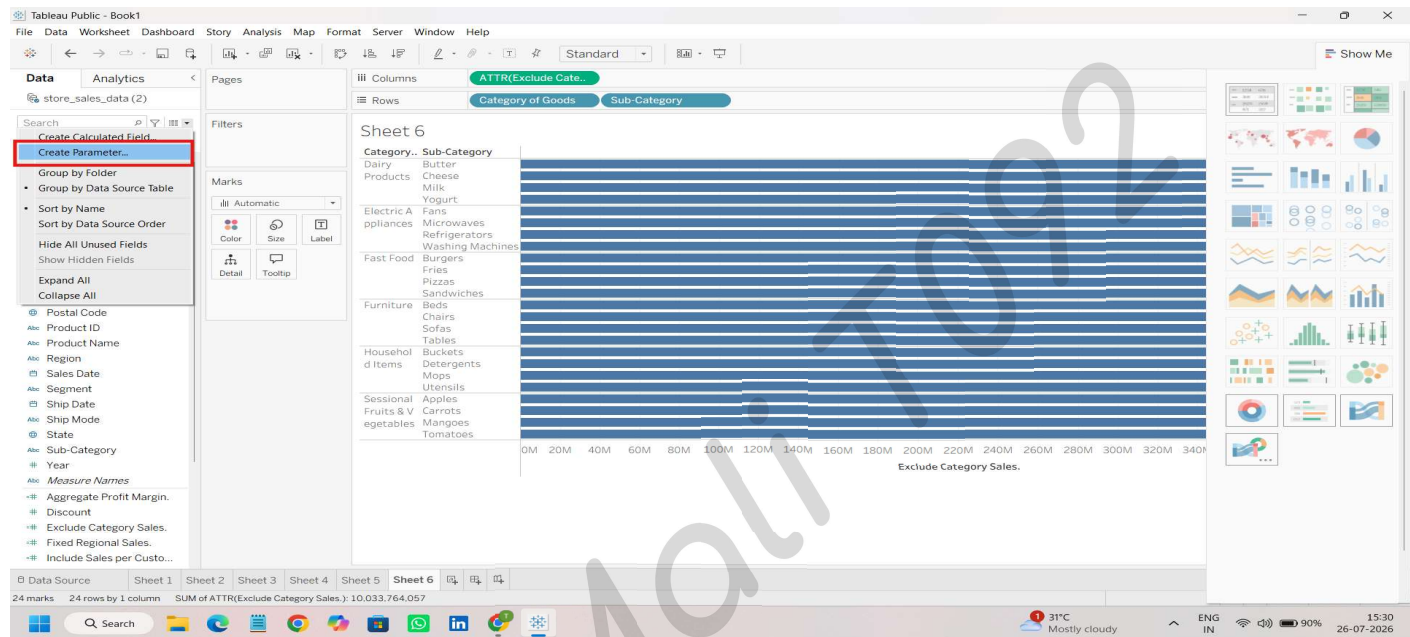


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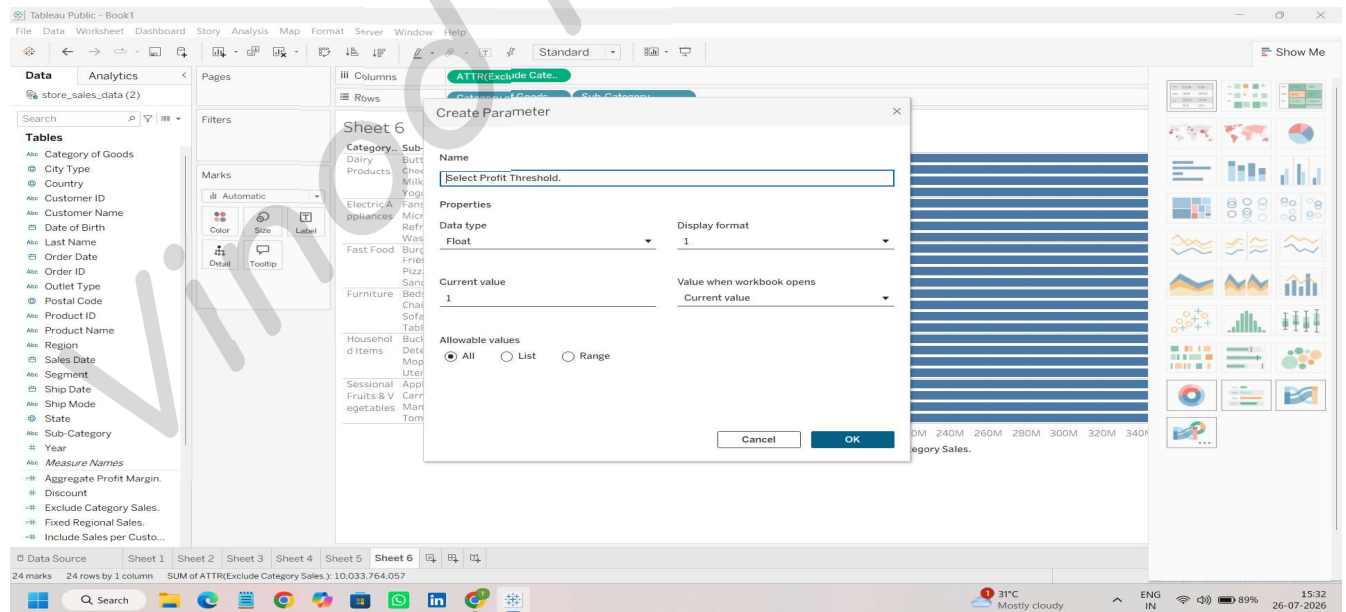
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F) Create and use parameters for dynamic analysis

Step 1:- Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.



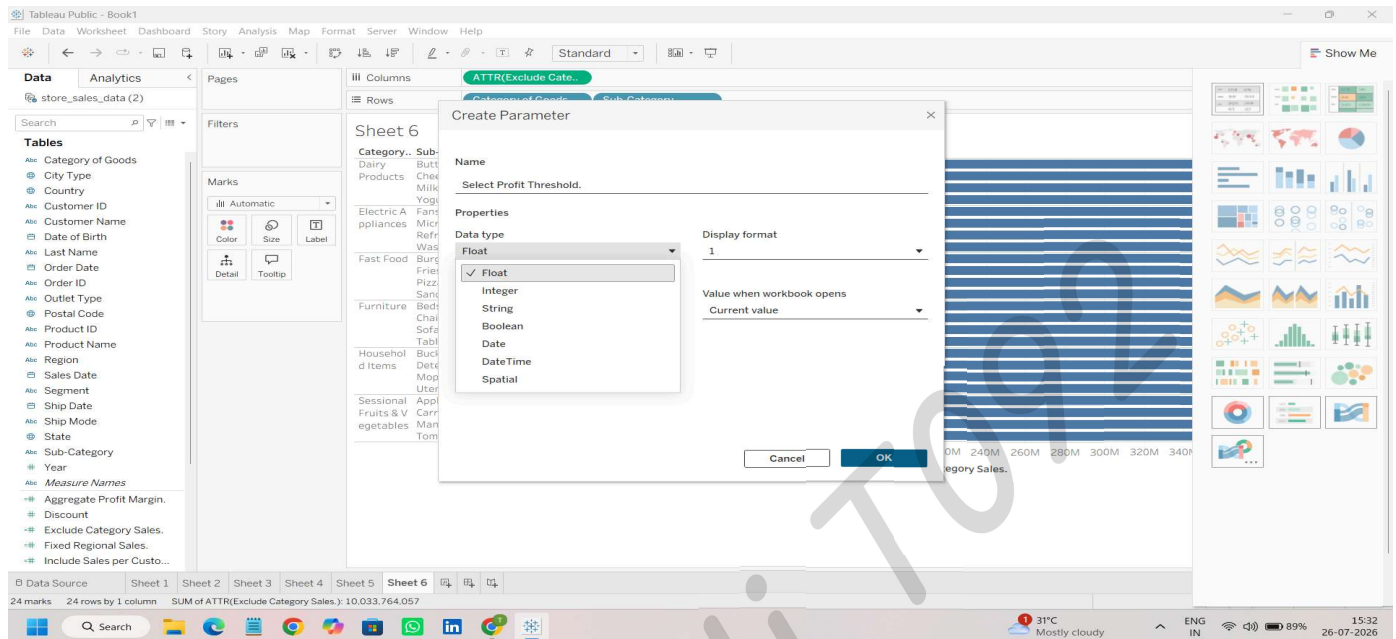
Step 2:- Name the parameter: Select Profit Threshold.



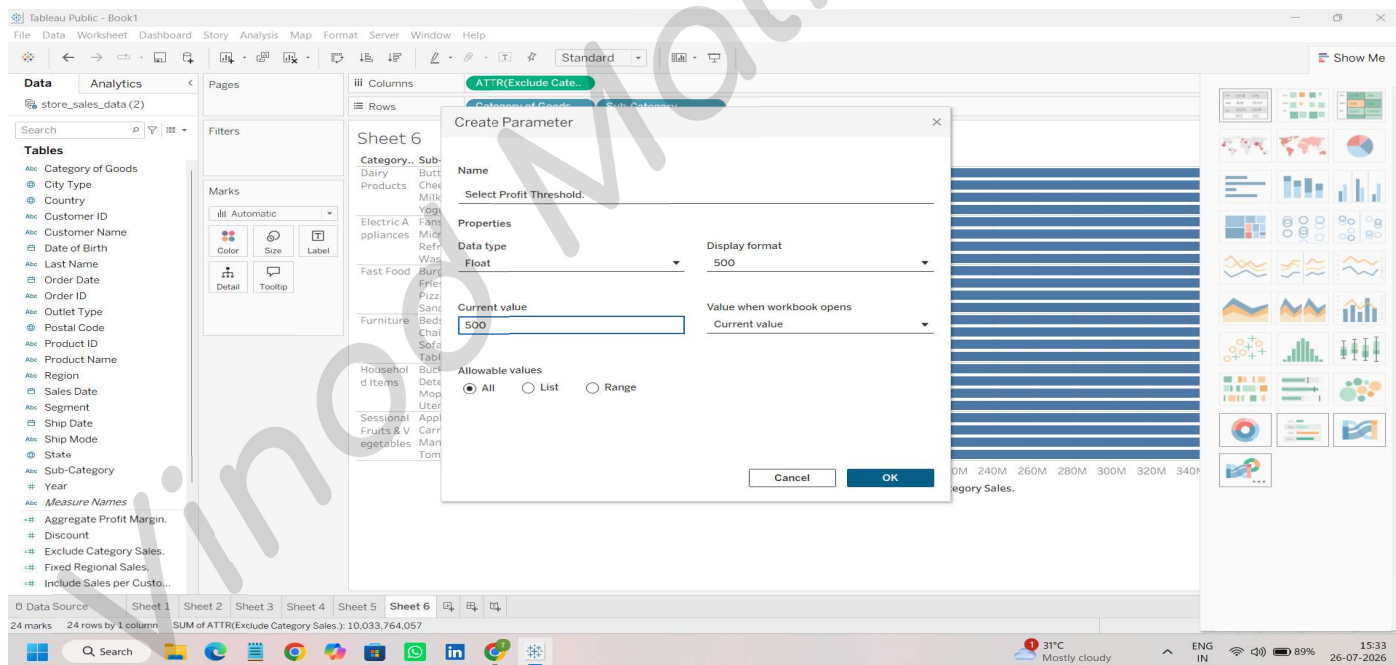
Step 3:- Set the Data Type to Float (or Integer).

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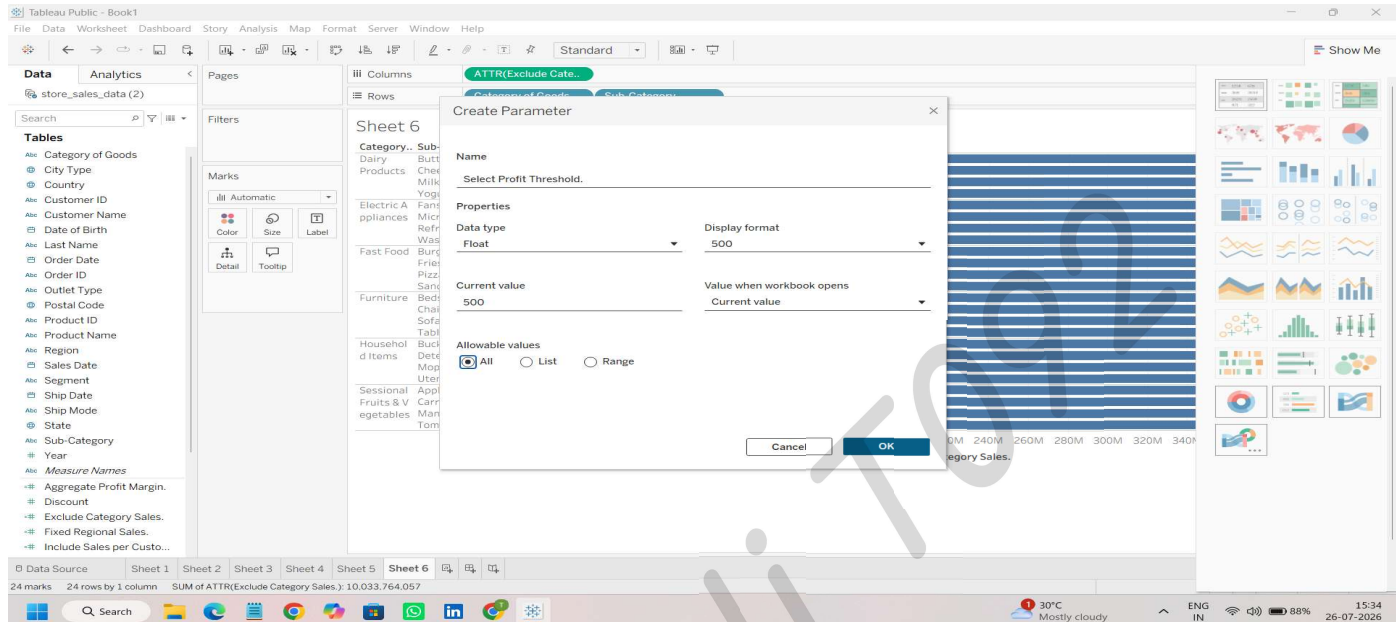
Step 4:- Set the Current Value to 500.



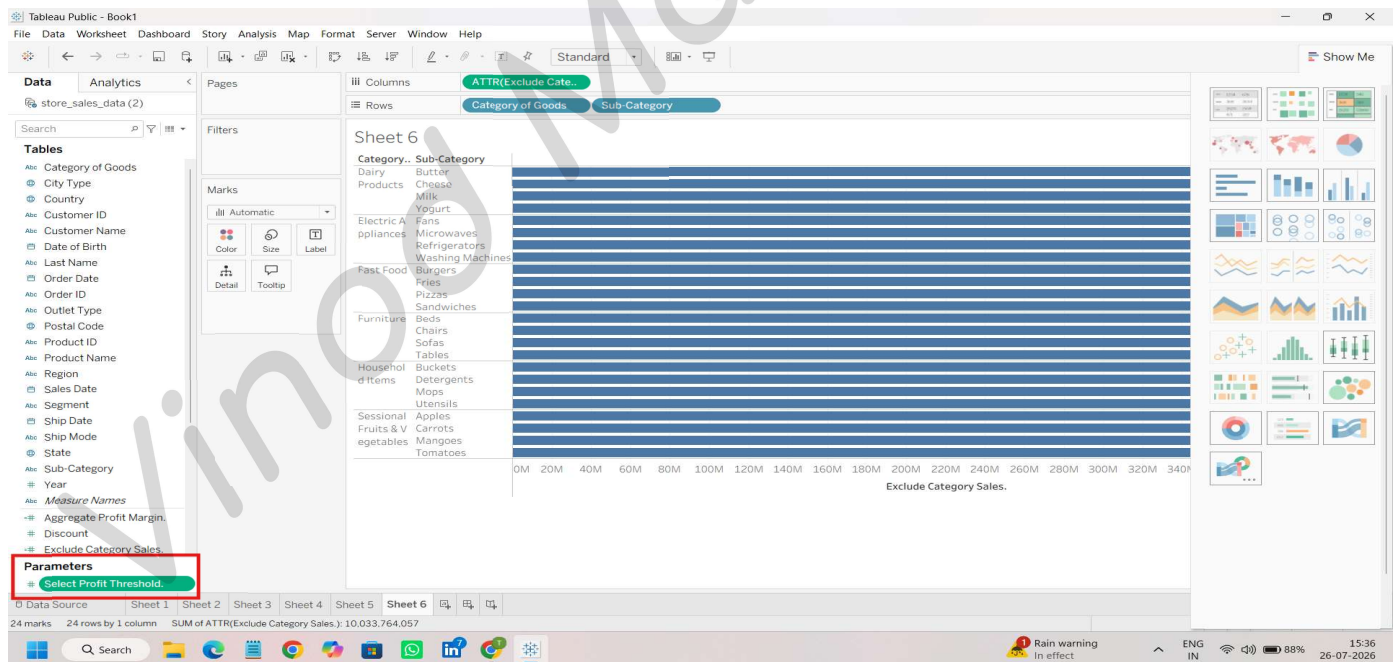
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Step 5:- Keep the Allowable values set to All and click OK.



Step 6:- Notice a new section titled Parameters appears at the bottom of your Data Pane, showcasing your new selector.

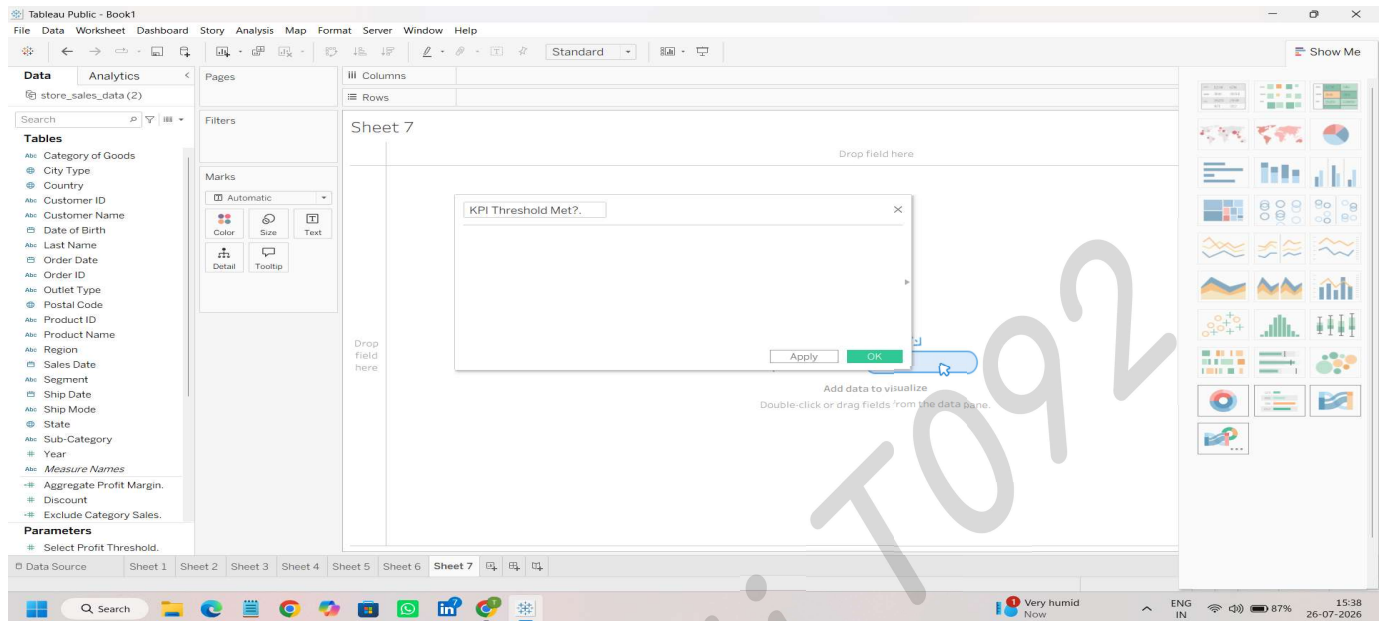


Step 8:- Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field... Name this field: KPI Threshold Met?

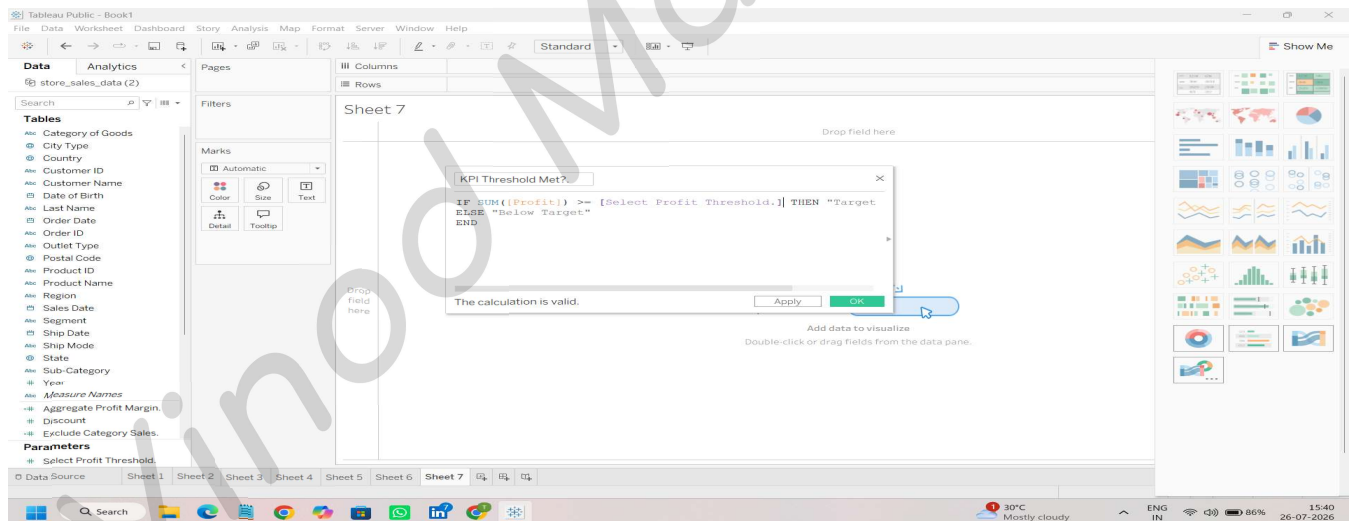
Name:- Vinod Manohar Mali

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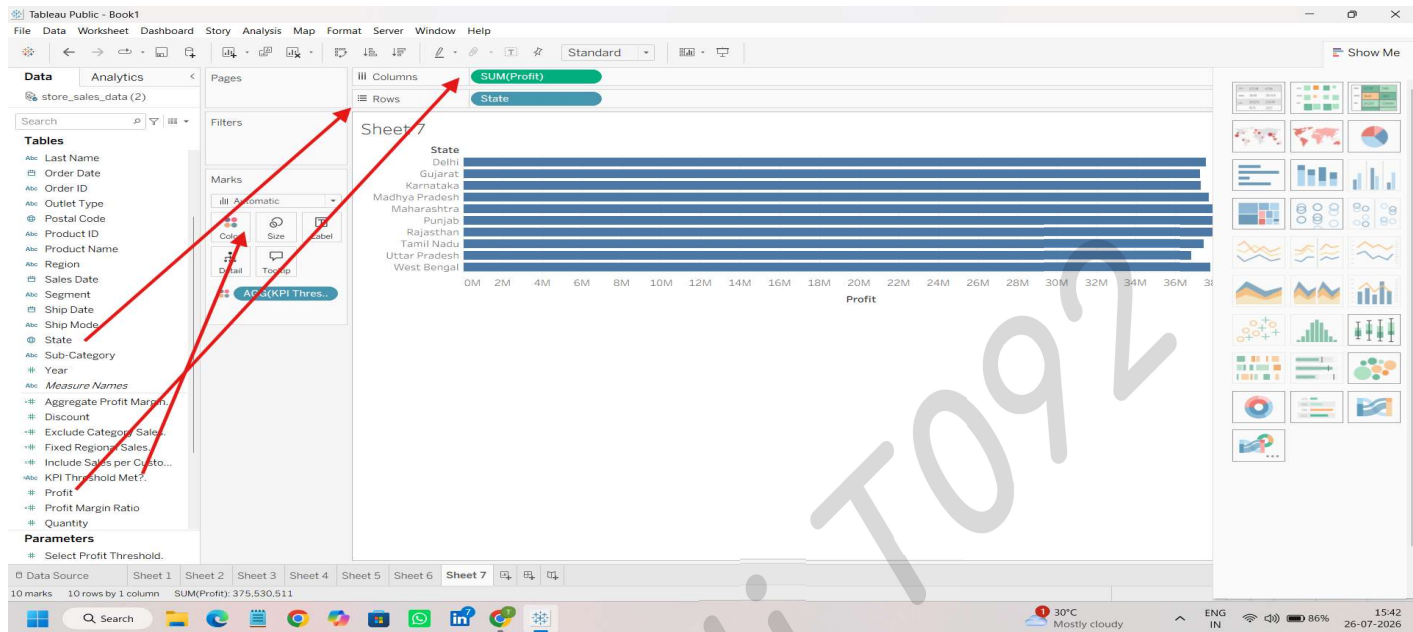
Step 9:- Input the following logical IF-THEN conditional statement:
IF SUM([Profit]) >= [Select Profit Threshold] THEN "Target Achieved"
ELSE "Below Target" END



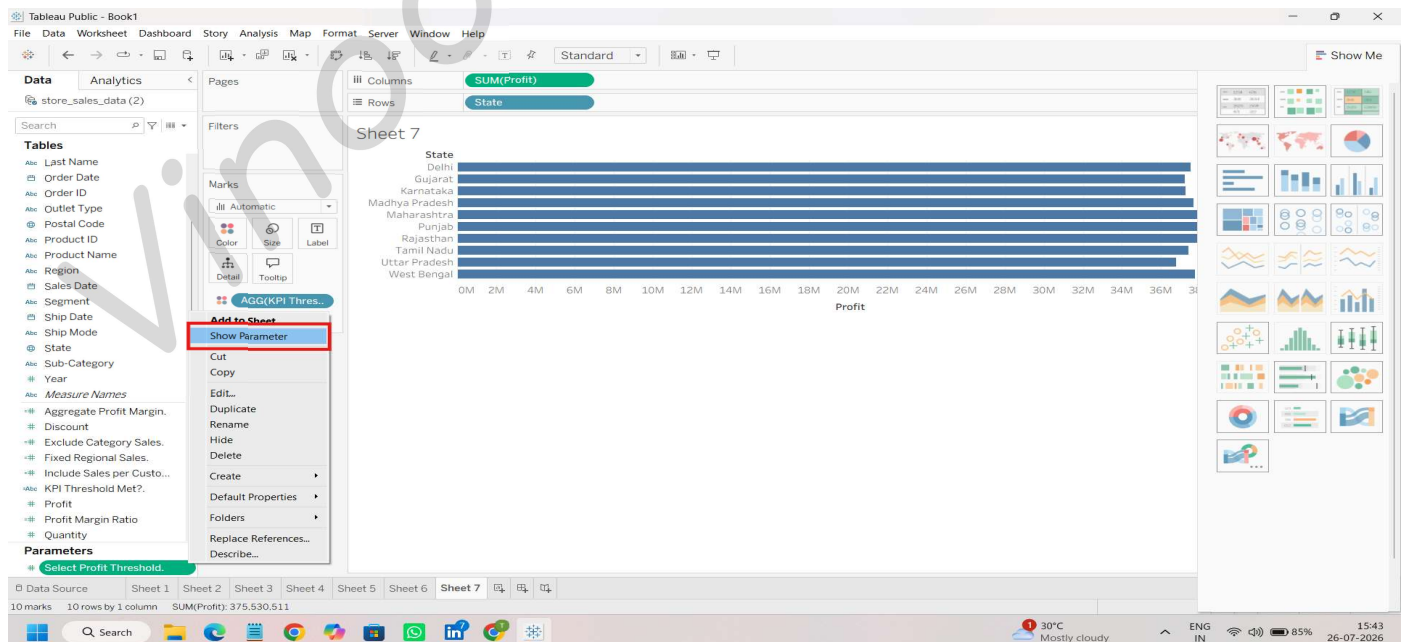
Step 10:- Open a new blank worksheet. Drag the State dimension from your Data Pane and drop it onto the Rows shelf. Drag the Profit measure and drop it onto the Columns shelf. Drag your newly created calculated field KPI Threshold Met? and drop it directly onto the Color tile in the Marks card.

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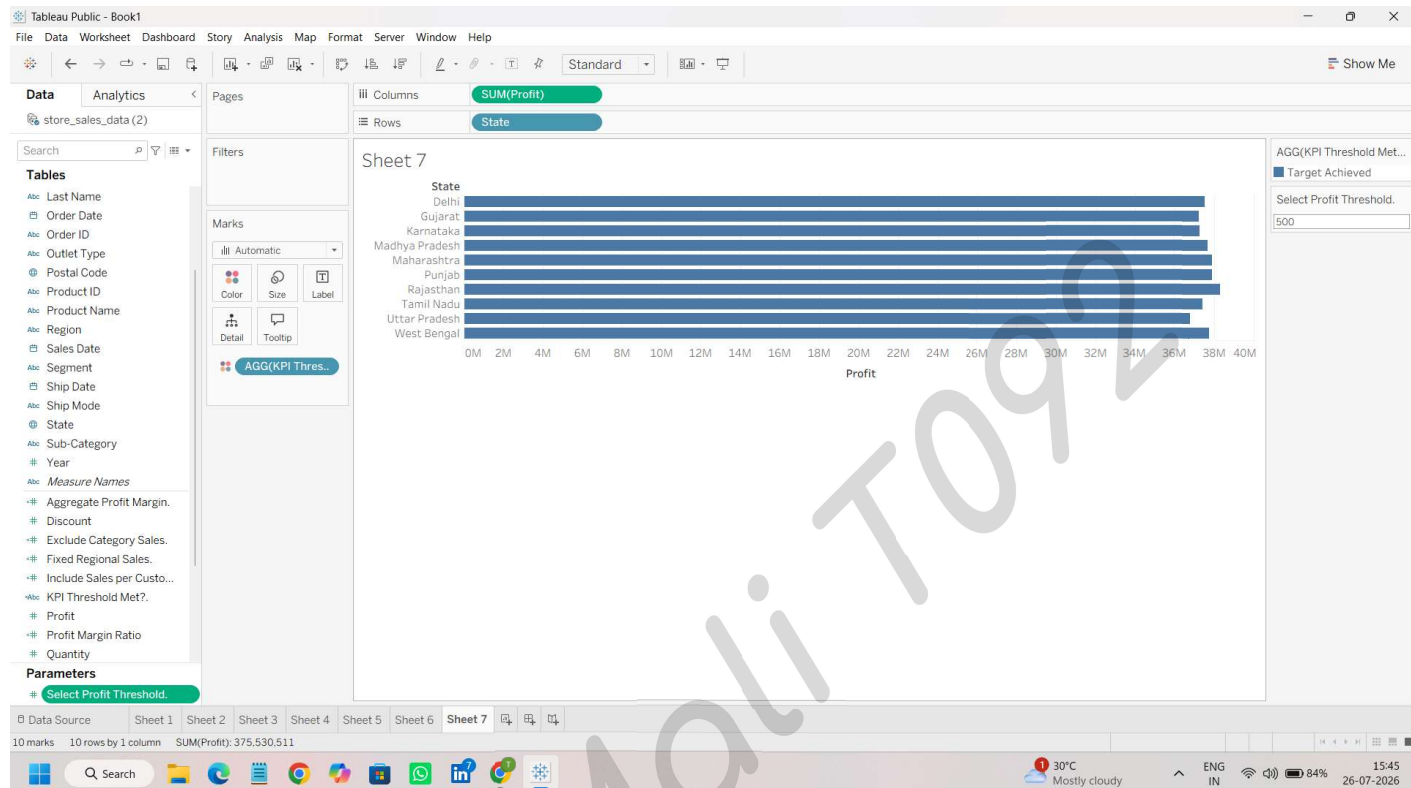


Step 11:- Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.



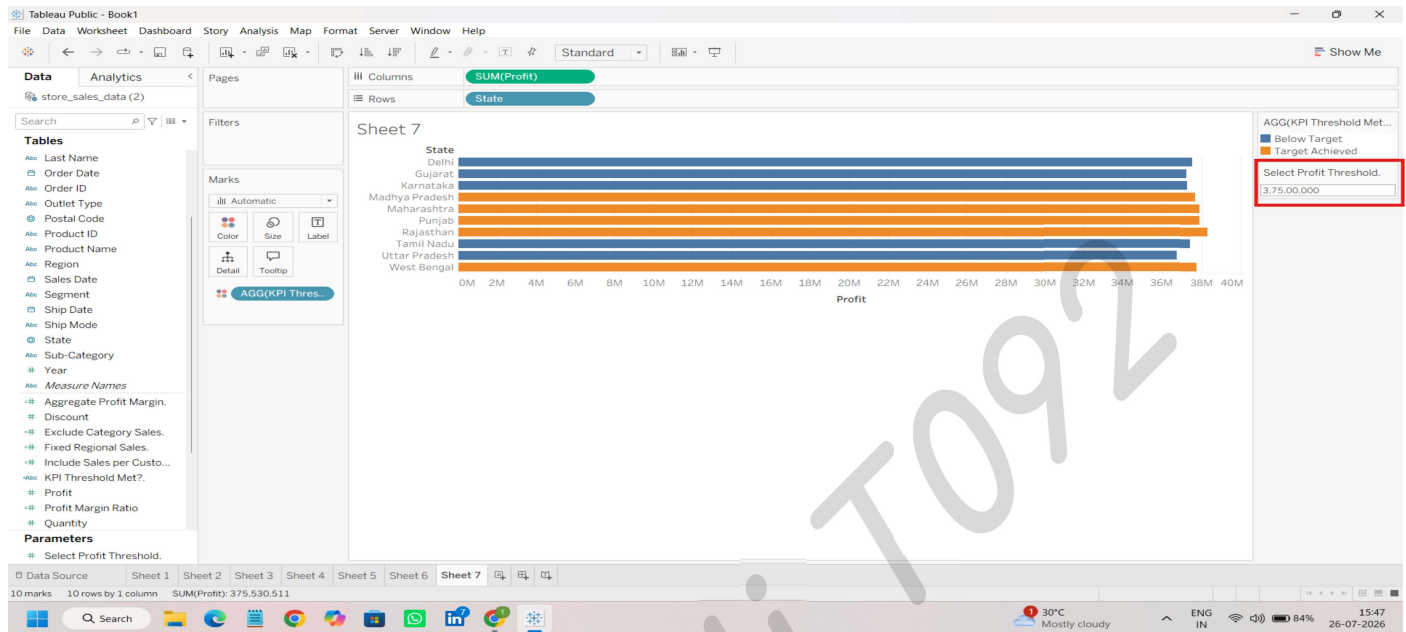
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Step 12:- A text box slider will slide into the upper right corner of your sheet canvas view. Type a low value like 100 or a high value like 5000 into that parameter box and hit Enter (NOTE: This is subjective to dataset in our case we typed 3,75,00,000 and the color changed instantly). The bars will instantly change color dynamically as states cross above or below the threshold you typed!

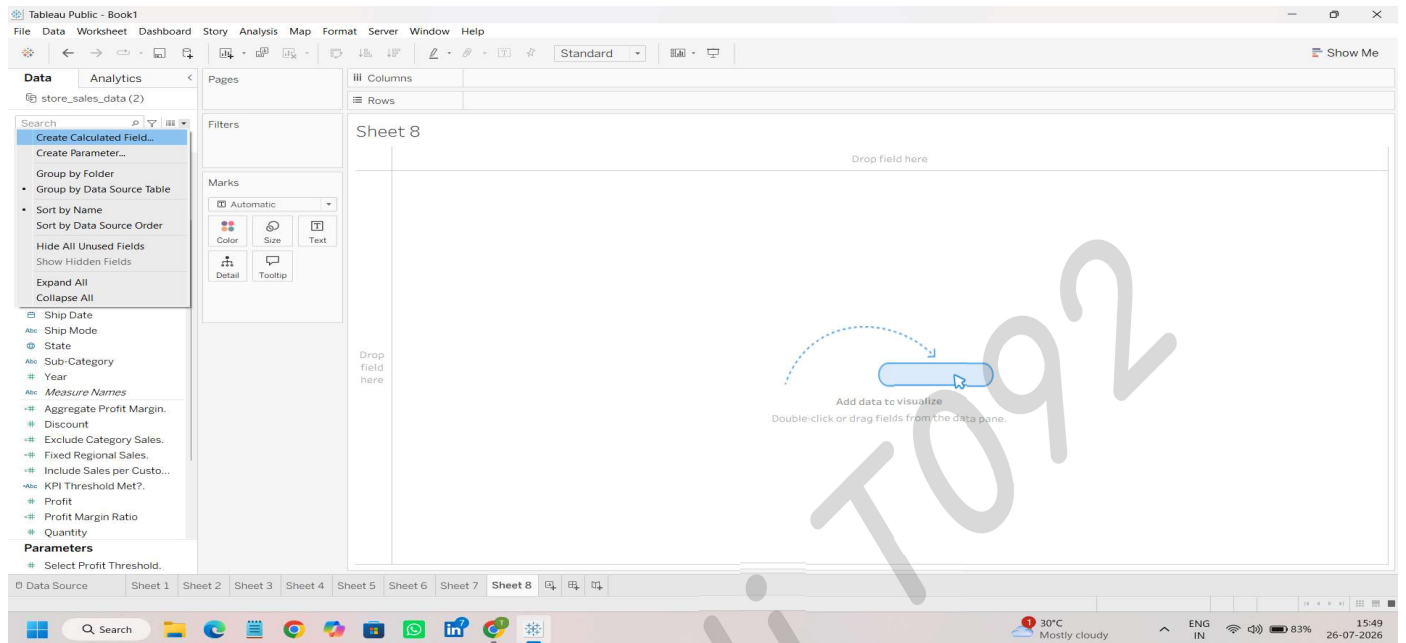
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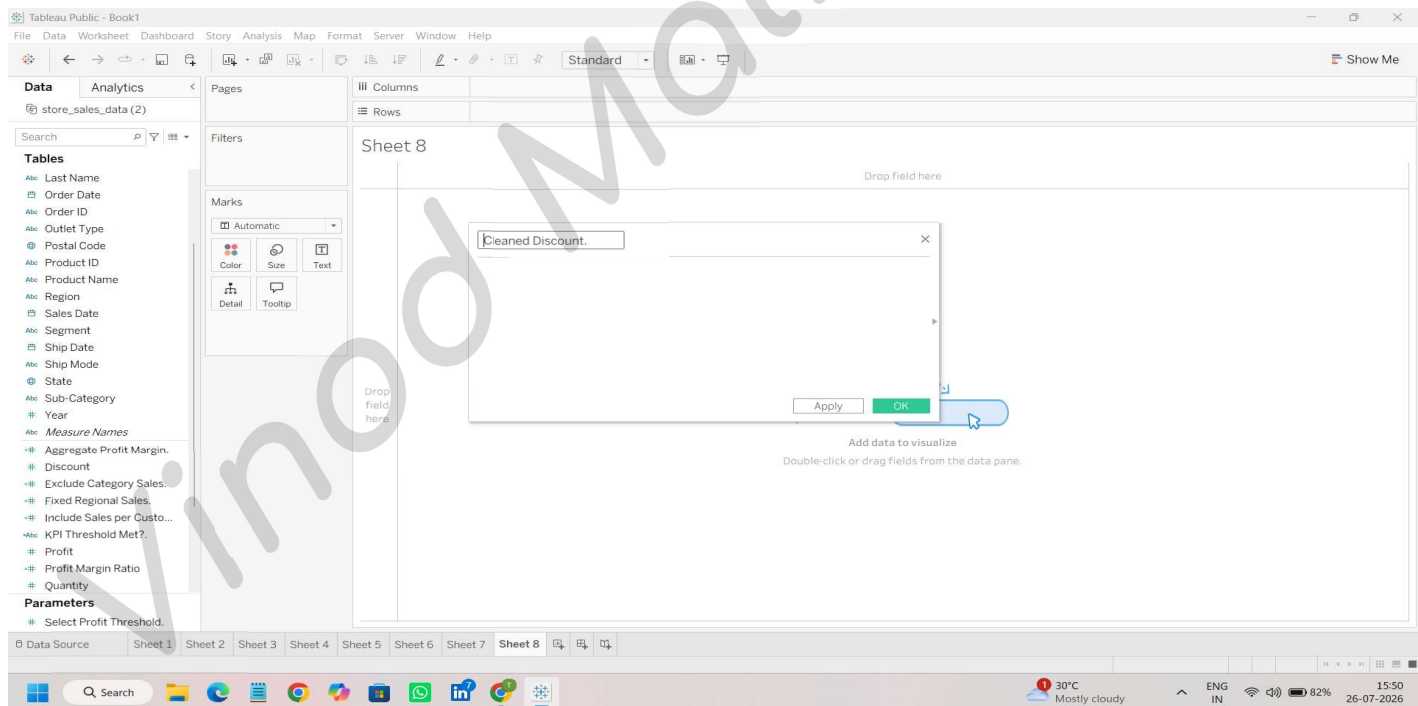
G) Apply calculations to fix data issues and enhance insights
Step 1:- Right-click in the blank white space of your Data Pane and select Create Calculated Field...

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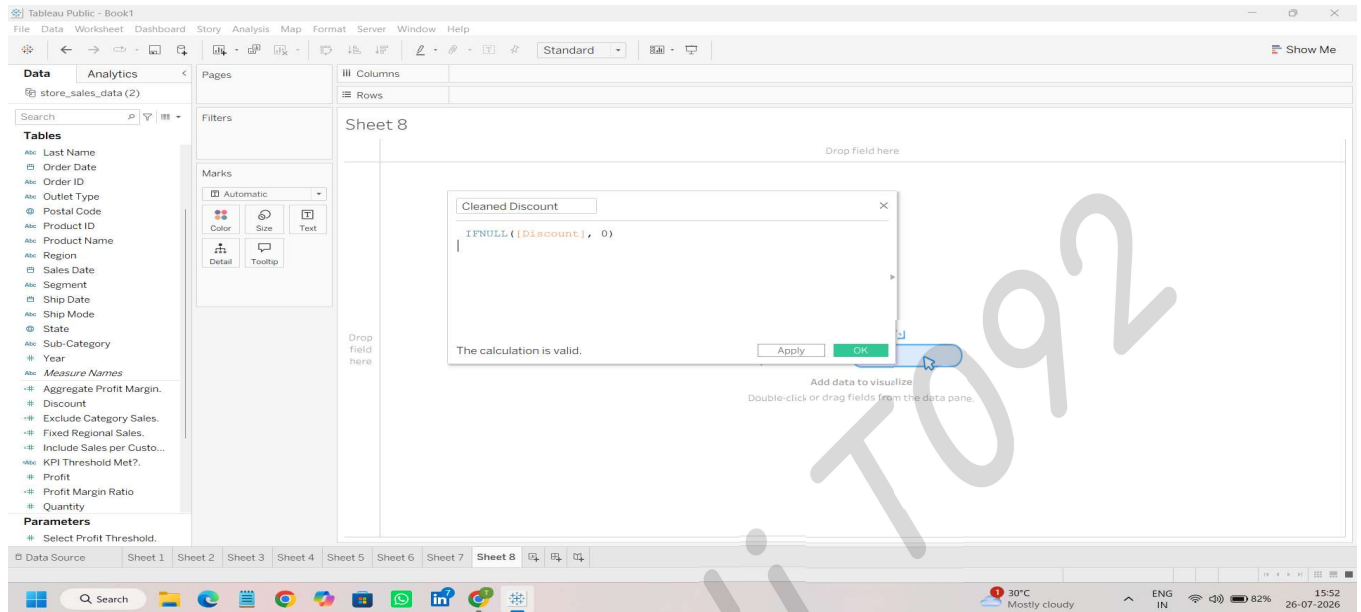
Step 2:- Name the field: Cleaned Discount.



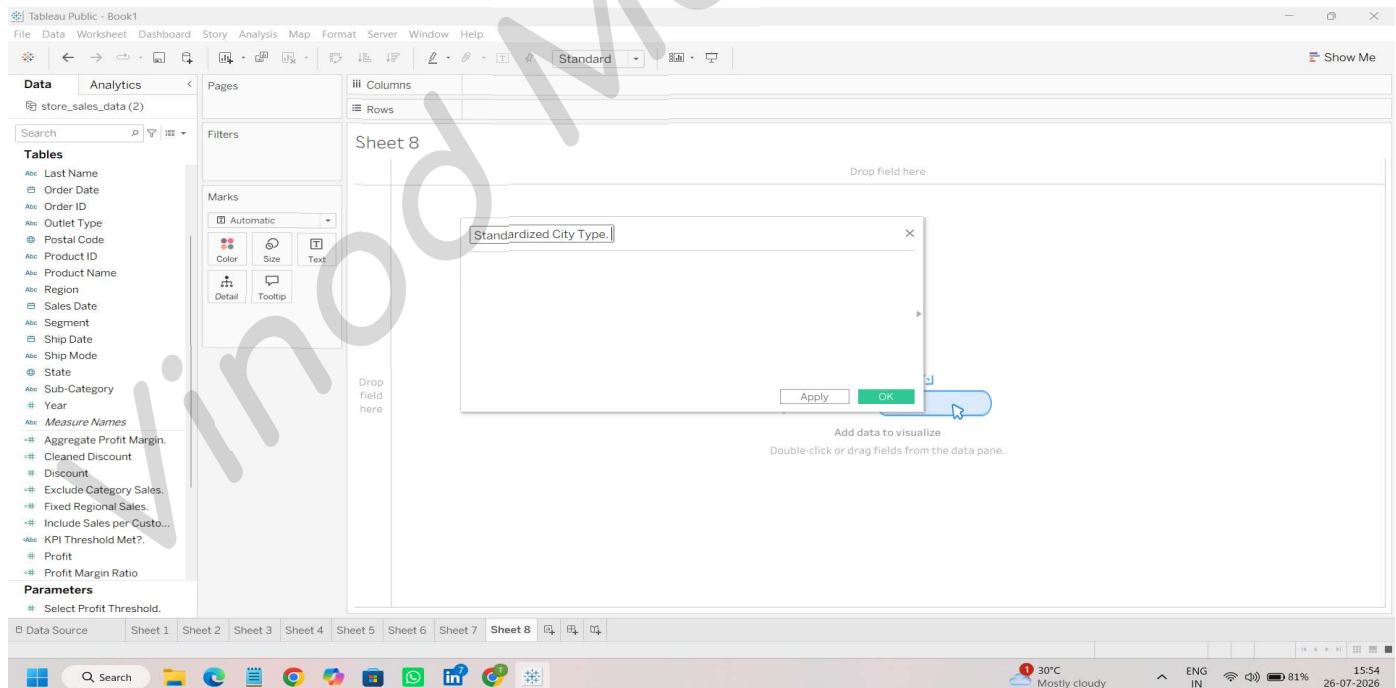
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Step 3:- Input the following expression: IFNULL([Discount], 0)

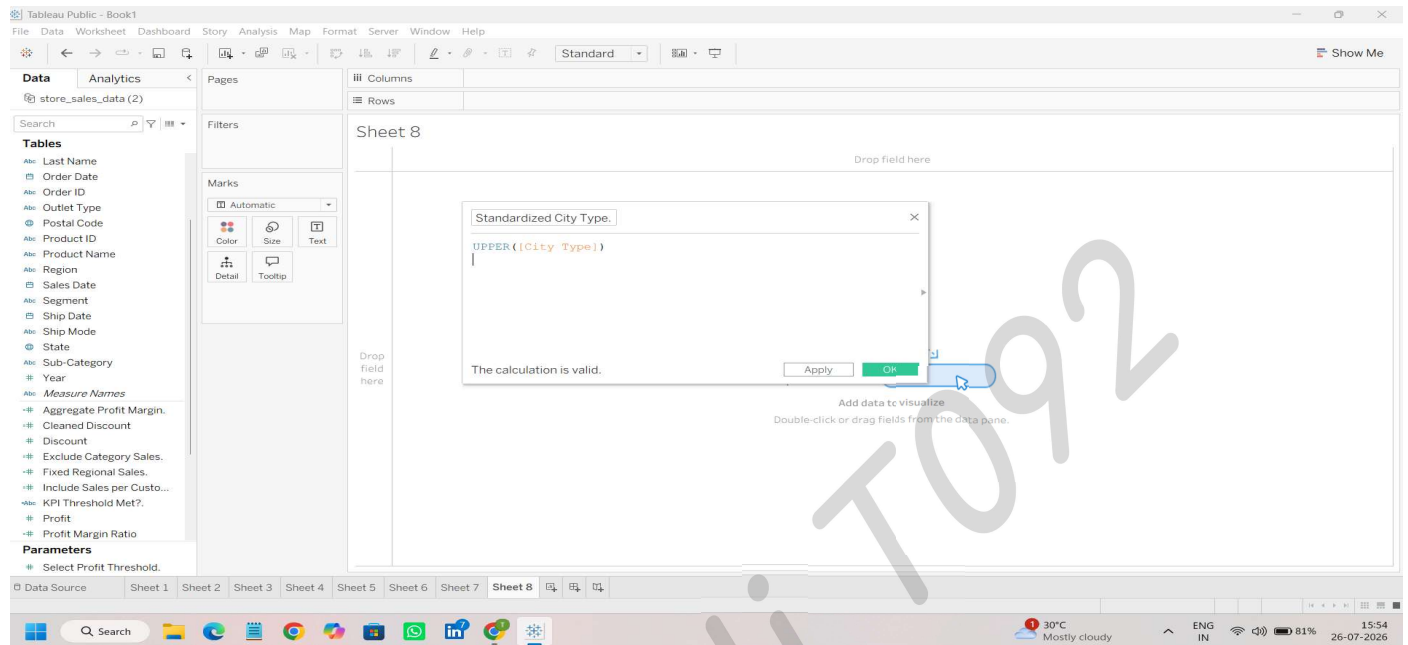


Step 4:- Create a new Calculated Field named Standardized City Type. Input the following text transformation expression: UPPER([City Type])

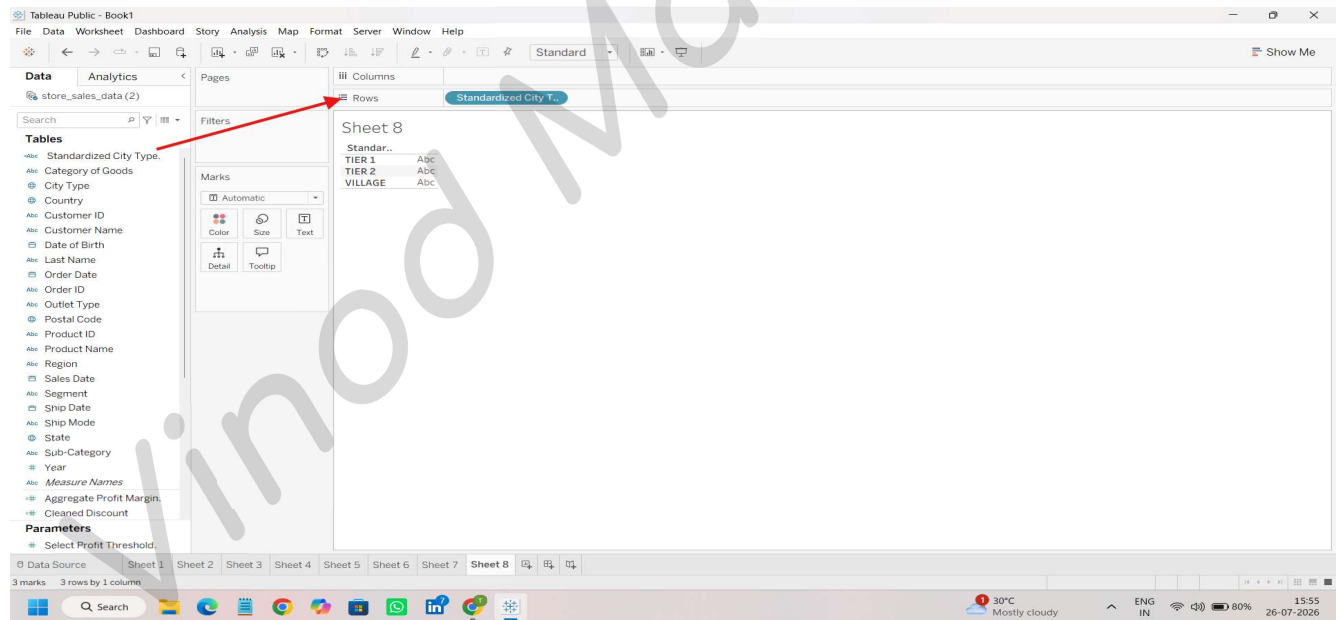


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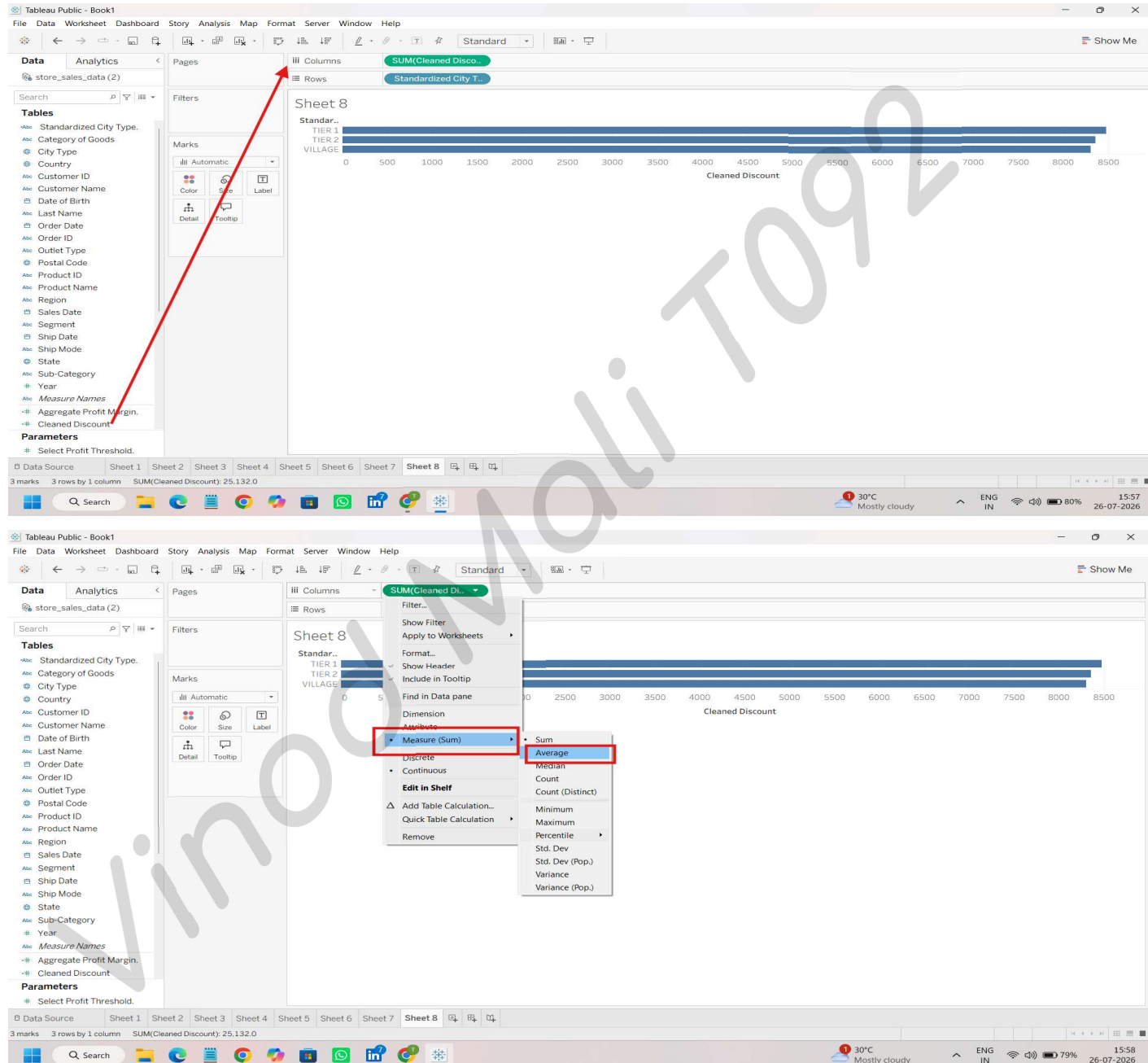
Step 5:- Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.



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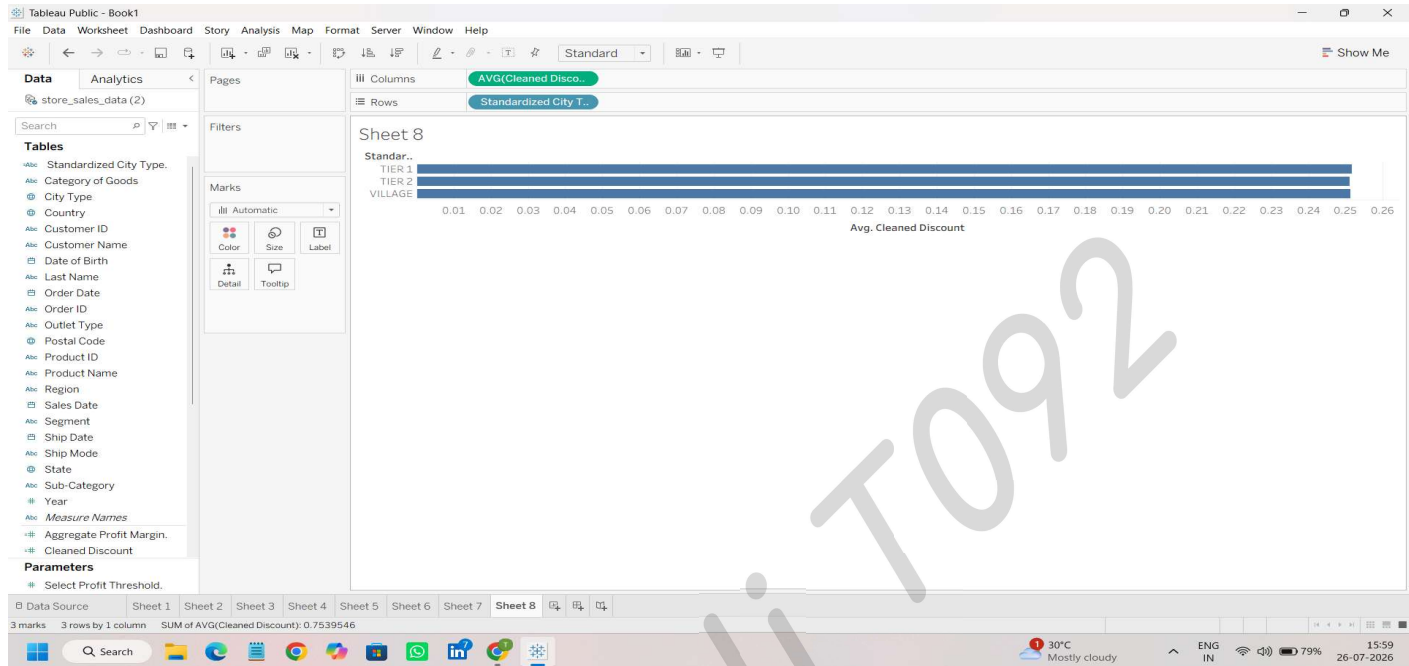
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Step 6:- Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by right-clicking the pill.



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